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CAMBRIDGE IELTS 17

HƯỚNG DẪN SỬ DỤNG SÁCH

ĐỐI TƯỢNG SỬ DỤNG SÁCH

Nhìn chung các bạn cần có mức độ từ vựng tương đương 5.5 trở lên (theo thang điểm 9 của IELTS), nếu không có thể sẽ gặp nhiều khó khăn trong việc sử dụng sách này.

CÁC BƯỚC SỬ DỤNG

CÁCH 1: LÀM TEST TRƯỚC, HỌC TỪ VỰNG SAU

Bước 1: Bạn in cuốn sách này ra. Nên in bìa màu để có thêm động lực học. Cuốn sách được thiết kế cho việc đọc trực tiếp, không phải cho việc đọc online nên bạn nào đọc online sẽ có thể thấy khá bất tiện khi tra cứu, đối chiếu từ vựng

Bước 2: Tìm mua cuốn Cambridge IELTS (Các cuốn mới nhất từ 8-16) của Nhà xuất bản Cambridge để làm. Hãy cẩn thận đừng mua nhầm sách lậu. Sách của nhà xuất bản Cambridge được tái bản tại Việt Nam thường có bìa và giấy dày, chữ rất rõ nét.

Bước 3: Làm một bài test hoặc passage bất kỳ trong bộ sách trên. Ví dụ passage 1, test 1 của Cambridge IELTS 13.

Bước 4: Đối chiếu với cuốn sách này, bạn sẽ lọc ra các từ vựng quan trọng cần học.
Ví dụ passage 1, test 1 của Cambridge IELTS 13, bài về Tourism New Zealand Website: Bạn sẽ thấy
4.1 Cột bên trái là bản text gốc, trong đó bôi đậm các từ học thuật - **academic word**
4.2 **Cột bên phải chứa các từ vựng này theo kèm định nghĩa (definition) hoặc từ đồng nghĩa (synonym)**

CÁCH 2: HỌC TỪ VỰNG TRƯỚC, ĐỌC TEST SAU

Bước 1: Bạn in cuốn sách này ra. Nên in bìa màu để có thêm động lực học. Cuốn sách được thiết kế cho việc đọc trực tiếp, không phải cho việc đọc online nên bạn nào đọc online sẽ có thể thấy khá bất tiện khi tra cứu, đối chiếu từ vựng

Bước 2: Đọc cột bên trái như đọc báo. Duy trì hàng ngày. Khi nào không hiểu từ nào thì xem nghĩa hoặc synonym của từ đó ở cột bên phải. Giai đoạn này giúp bạn phát triển việc đọc tự nhiên, thay vì đọc theo kiểu làm test. Bạn càng hiểu nhiều càng tốt. Cố gắng nhớ từ theo ngữ cảnh.

Bước 3: Làm một bài test hoặc passage bất kỳ trong bộ sách Cambridge IELTS. Ví dụ bạn đọc xong cuốn Boost your vocabulary 13 này thì có thể quay lại làm các test trong cuốn 10 chẳng hạn. **Làm test xong thì cố gắng phát hiện các từ đã học** trong cuốn 13. Bạn nào có khả năng ghi nhớ tốt chắc chắn sẽ gặp lại rất nhiều từ đã học. Bạn nào có khả năng ghi nhớ vừa phải cũng sẽ gặp lại không ít từ.

Bước 4: Đọc cuốn Boost your vocabulary tương ứng với test bạn vừa làm. Ví dụ trong cuốn Boost your vocabulary 10.

Tóm lại, mình ví dụ 1 chu trình đầy đủ theo cách này

B1. Đọc **hiểu** và học từ cuốn Boost your vocabulary 13

B2. Làm test 1 trong cuốn Boost your vocabulary 10

B3. Đọc **hiểu** và học từ cuốn Boost your vocabulary 10 & tìm các từ lặp lại mà bạn đã đọc trong cuốn Boost your vocabulary 13

TEST 1

READING PASSAGE 1

In the first half of the 1800s, London's population grew at an **astonishing** rate, and the central area became increasingly **congested**. In addition, the **expansion** of the overground railway network resulted in more and more passengers arriving in the capital. However, in 1846, a Royal Commission decided that the railways should not be allowed to enter the City, the capital's historic and business centre. The result was that the overground railway **stations** formed a **ring** around the City. The area within consisted of poorly built, overcrowded **slums** and the streets were full of **horse-drawn** traffic. Crossing the City became a nightmare. It could take an hour and a half to travel 8 km by horse-drawn **carriage** or bus. **Numerous schemes** were **proposed** to **resolve** these problems, but few succeeded.	**railway**= a system of tracks that trains travel along **astonishing**= surprising, shocking, astounding **congested**= overcrowded, crammed, blocked **expansion**= extension, growth, enlargement **station**= a building and the surrounding area where buses or trains stop for people to get on or off **ring**= circle, loop, sphere **slum**= a very poor and crowded area, especially of a city **horse-drawn**= a horse-drawn vehicle is pulled by a horse. **carriage**= a vehicle with four wheels that is usually pulled by horses and was used mainly in the past **numerous**= many, plentiful, various **scheme**= plan, method, idea **propose**= suggest, offer, recommend **resolve**= solve, sort out, settle

Amongst the most **vocal advocates** for a solution to London's traffic problems was Charles Pearson, who worked as a **solicitor** for the City of London. He saw both social and economic advantages in building an underground railway that would **link** the overground railway stations together and **clear** London slums at the same time. His idea was to **relocate** the poor workers who lived in the **inner-city** slums to newly **constructed suburbs**, and to provide cheap rail travel for them to get to work. Pearson's ideas gained support amongst some businessmen and in 1851 he **submitted** a plan to **Parliament**. It was **rejected**, but **coincided** with a **proposal** from another group for an underground connecting **line**, which Parliament **passed**.	**vocal**= outspoken, loud, forceful **advocate**= supporter, promoter, activist **solicitor**= a type of lawyer in Britain and Australia **link**= connect, join, bring together **clear**= tidy up, clear out, empty **relocate**= move, displace, change place **inner-city**= in the central part of a city where there are often problems because people are poor and there are few jobs and bad houses **construct**= build, make, create **suburb**= an area on the edge of a large town or city **submit**= present, offer, suggest **parliament**= the group of people who make the laws for their country **reject**= refuse, decline, deny **coincide**= happen together, overlap, match **proposal**= suggestion, request, offer **line**= a railway track **pass**= accept, permit, approve
The two groups **merged** and established the Metropolitan Railway Company in August 1854. The company's plan was to construct an underground railway line from the Great Western Railway's (GWR) station at Paddington to the edge of the City at Farringdon Street - a distance of almost 5 km. The organisation had difficulty in raising the funding for such a **radical** and expensive scheme, not least because of the **critical** articles printed by the **press**. **Objectors** argued that the **tunnels** would **collapse** under the weight of traffic overhead, buildings would be shaken and passengers would be **poisoned** by the **emissions** from the train **engines**. However, Pearson and his partners **persisted**.	**merge**= combine, join together, team up **radical**= extreme, far-out, progressive **critical**= disapproving, fault-finding, unfavorable **press** = media, newspapers, journalists **objector**= opponent, skeptic, critic **tunnel**= a long passage under or through the ground **collapse**= breakdown, fall to pieces, fail **poison**= harm, infect, injure **emission**= exhaust fumes **engine**= machine, piece of equipment, mechanism **persist**= continue, carry on, stick with
The GWR, aware that the new line would finally enable them to run trains into the **heart** of the City, invested almost £250,000 in the scheme. **Eventually**, over a five-year period, £1m was **raised**. The chosen **route** ran beneath existing main roads to minimise the **expense** of **demolishing** buildings. **Originally scheduled** to be completed in 21 months, the construction of the underground line took three years. It was built just below street level using a technique known as 'cut and cover'. A **trench** about ten metres wide and six metres deep was dug, and the **sides temporarily** held up with **timber** beams. Brick walls were then constructed, and finally a brick **arch** was added to create a tunnel. A two-metre-deep layer of soil was laid on top of the tunnel and the road above rebuilt.	**heart**= center, core, middle **eventually**= finally, in the end, ultimately **raise**= to raise money is to succeed in getting it **route**= way, road, track **expense**= cost, payment, expenditure **demolish**= destroy, ruin, wreck **schedule**= arrange, plan, organize **originally**= firstly, in the beginning, initially **trench**= a narrow channel dug into the ground **side**= a flat outer surface of an object, especially one that is not the top, the bottom, the front, or the back **temporarily**= in the short term, briefly, provisionally **beam**= a long, thick piece of wood, metal, or concrete, especially used to support weight in a building or other structure **timber**= wood, logs, kindling **arch**= a structure, consisting of a curved top on two supports, that holds the weight of something above it
The Metropolitan line, which opened on 10 January 1863, was the world's first underground railway. On its first day, almost 40,000 passengers were **carried** between Paddington and Farringdon, the journey taking about 18 minutes. By the end of the Metropolitan's first year of operation, 9.5 million journeys had been made.	**carry**= transport, bring, transfer
Even as the Metropolitan began operation, the first **extensions** to the line were being **authorised**; these were built over the next five	**extension**= lengthening, expansion, increase **authorise**= approve, permit, give permission

years, reaching Moorgate in the east of London and Hammersmith in the west. The original plan was to pull the trains with **steam locomotives**, using **firebricks** in the **boilers** to provide steam, but these engines were never **introduced**. Instead, the line used specially designed locomotives that were fitted with **water tanks** in which steam could be **condensed**. However, smoke and **fumes** remained a problem, even though **ventilation shafts** were added to the tunnels.	**steam locomotive**= a vehicle with an engine powered by steam, used for pulling trains **firebrick**= a type of brick that is not damaged by high temperatures **boiler**= a device that heats water **introduce**= begin, launch, start **water tank**= a large container for collecting and storing water **condense**= to change or make something change from a gas to a liquid or solid state **fume**= gas, smog, emission **ventilation** = air circulation, freshening, airing **shaft**= a long passage through a building or through the ground
Despite the extension of the underground railway, by the 1880s, **congestion** on London's streets had become worse. The problem was partly that the existing underground lines formed a **circuit** around the centre of London and extended to the suburbs, but did not cross the capital's centre. The 'cut and cover' method of construction was not an option in this part of the capital. The only **alternative** was to **tunnel** deep underground.	**congestion**= overcrowding, jamming, blocking **circuit**= route, path, track **alternative**= another possibility, substitute, replacement **tunnel**= dig, excavate, burrow
Although the technology to create these tunnels existed, steam locomotives could not be used in such a **confined** space. It wasn't until the development of a **reliable** electric **motor**, and a **means** of transferring power from the **generator** to a moving train, that the world's first deep-level electric railway, the City & South London, became possible. The line opened in 1890, and ran from the City to Stockwell, south of the River Thames. The trains were made up of three **carriages** and driven by electric engines. The carriages were narrow and had tiny windows just below the roof because it was thought that passengers would not want to look out at the tunnel walls. The line was not without its problems, mainly caused by an unreliable power supply. Although the City & South London Railway was a great **technical** achievement, it did not make a profit. Then, in 1900, the Central London Railway, known as the 'Tuppenny Tube', began operation using new electric locomotives. It was very popular and soon afterwards new railways and extensions were added to the growing **tube** network. By 1907, the heart of today's Underground system was **in place**.	**confined**= small, cramped, enclosed **reliable**= trustworthy, dependable, unfailing **motor**= a device that changes electricity or fuel into movement and makes a machine work **means**= way, method, measure **generator**= power producer **carriage**= any of the separate parts of a train in which the passengers sit **technical**= mechanical, industrial, scientific **tube**= London's underground train system **in place**=ready, ripe, primed

TEST 1

READING PASSAGE 2

A. Stadiums are among the oldest forms of urban **architecture**: **vast** stadiums where the public could watch sporting events were at the centre of western city life as far back as the ancient Greek and Roman **Empires**, well before the **construction** of the great **medieval cathedrals** and the **grand** 19th- and 20th-century railway **stations** which **dominated** urban **skylines** in later eras.

Today, however, stadiums are **regarded** with growing **scepticism**. Construction costs can **soar** above £1 billion, and stadiums finished for **major** events such as the Olympic Games or the FIFA World Cup have **notably fallen into** disuse and disrepair.

But this need not be **the case**. History shows that stadiums can **drive** urban development and **adapt** to the culture of every **age**. Even today, **architects** and planners are finding new

architecture= design, building, style
vast= huge, enormous, massive
empire= a group of countries ruled by a single person, government, or country
construction= building, creation, development
medieval= of or from the middle ages (= the period in the past from about 500 to 1500)
cathedral= a very large, usually stone, building for christian worship
grand= large, huge, massive
station= depot, terminal, stop
dominate= to be the largest or most noticeable part of something
skyline= the shape of objects against the sky, esp. buildings in a city
regard= think, consider, deem
scepticism= disbelief, doubt, uncertainty
soar= rise, escalate, rocket
major= most important, main, key
notably= especially, particularly, remarkably
fall into= to gradually get into a particular condition, especially to get into a bad condition

(not) the case= (not) true
drive= push, force, propel
adapt= fit, modify, adjust
age= period, time, era
architect= designer, engineer, builder

ways to adapt the **mono-functional** sports **arenas** which became **emblematic** of modernisation during the 20th century.	**mono-functional**= having a single function **arena**= sports ground, stadium, pitch **emblematic**= symbolic, representative, characteristic
B. The amphitheatre* of Aries in southwest France, with a **capacity** of 25,000 **spectators**, is perhaps the best example of just how **versatile** stadiums can be. Built by the Romans in 90 AD, it became a **fortress** with four towers after the fifth century, and was then transformed into a village containing more than 200 houses. With the growing **interest** in **conservation** during the 19th century, it was **converted** back into an arena for the **staging** of bullfights, **thereby** returning the structure to its original use as a **venue** for public **spectacles**.	**capacity**= volume, size, space **spectator**= viewer, watcher, observer **versatile**= flexible, adaptable, multipurpose **fortress**= a large, strong building or group of buildings that can be defended from attack **interest**= concern, attention, notice **conservation**= protection, preservation, maintenance **convert**= change, switch, alter **staging**= performance, presentation, production **thereby**= so, thus, in that way **venue**= site, location, setting **spectacle**= event, performance, display
Another example is the **imposing** arena of Verona in northern Italy, with space for 30,000 spectators, which was built 60 years before the Aries amphitheatre and 40 years before Rome's famous Colosseum. It has **endured** the centuries and is currently considered one of the world's **prime** sites for opera, thanks to its **outstanding acoustics**.	**imposing**= impressive, striking, magnificent **endure**= last, survive, persist **prime**= excellent, first-rate, top-notch **outstanding**= wonderful, excellent, exceptional **acoustic**= sound, audio, auditory
C. The area in the centre of the Italian town of Lucca, known as the Piazza dell'Anfiteatro, is yet another impressive example of an amphitheatre becoming **absorbed** into **the fabric of** the city. The site **evolved** in a similar way to Aries and was **progressively** filled with buildings from the Middle Ages until the 19th century, variously used as houses, a salt **depot** and a prison. But rather than reverting to an arena, it became a market square, designed by Romanticist architect Lorenzo Nottolini. Today, the **ruins** of the amphitheatre remain **embedded** in the various shops and **residences** surrounding the public square.	**absorb**= incorporate, merge, integrate **the fabric of**= the structure or parts of something **evolve**= grow, progress, develop **progressively**= gradually, little by little, with time **depot**= storehouse, warehouse, storage area **ruin**= debris, wreckage, remains **embed**= incorporate, lodge, fix **residence**= a home
D. There are many similarities between modern stadiums and the ancient amphitheatres **intended** for games. But some of the flexibility was lost at the beginning of the 20th century, as stadiums were developed using new products such as steel and **reinforced concrete**, and **made use of** bright lights for night-time matches.	**intend**= designate, aim, plan **reinforced concrete**= concrete that contains metal rods to make it stronger **make use of**= use, utilize, exploit
Many such stadiums are situated in suburban areas, designed for sporting use only and surrounded by parking lots. These factors mean that they may not be as **accessible** to the **general public,** require more energy to run and contribute to urban heat.	**accessible**= available, nearby, easy to get to **general public**= population, citizens, ordinary people
E. But many of today's most **innovative** architects see **scope** for the stadium to help improve the city. Among the current strategies, two seem to be having **particular** success: the stadium as an urban **hub**, and as a **power plant**.	**innovative**= modern, novel, groundbreaking **scope**= opportunity, possibility, chance **particular**= specific, exact, certain **hub**= the central or main part of something where there is most activity **power plant**= a factory where electricity is produced

There's a growing trend for stadiums to be **equipped** with public spaces and services that serve a function beyond sport, such as hotels, **retail outlets**, **conference** centres, restaurants and bars, children's playgrounds and green space. Creating mixed-use developments such as this **reinforces compactness** and multi-functionality, making more efficient use of land and helping to **regenerate** urban spaces.

equip= provide, give, furnish
retail outlet= a store that sells goods to the public
conference= meeting, seminar, discussion
reinforce= strengthen, bolster, support
compactness= neatness, smallness, trimness
regenerate= renew, redevelop, restart

This **opens** the space **up to** families and a wider **cross-section** of society, instead of **catering** only to **sportspeople** and **supporters**. There have been many examples of this in the UK: the mixed-use facilities at Wembley and Old Trafford have become a **blueprint** for many other stadiums in the world.

open sth up to= to make something available
cross-section=representation, sample
cater= serve, provide for, accommodate
sportspeople= athlete, sports player
supporter= fan, follower, enthusiast
blueprint=prototype, example

F. The phenomenon of stadiums as power stations has **arisen from** the idea that energy problems can be overcome by **integrating interconnected** buildings **by means of** a smart **grid**, which is an electricity supply network that uses digital communications technology to **detect** and react to local changes in **usage**, without **significant** energy losses. Stadiums are ideal for these purposes, because their **canopies** have a large surface area for fitting **photovoltaic panels** and rise high enough (more than 40 metres) to make use of **micro** wind **turbines.**

arise from= stem from, result from, develop out of
integrate= mix, add, combine
interconnected= connected, joined, interrelated
by means of= by, via, using
grid= network, net, web
detect= discover, notice, identify
usage= the way something is treated or used
significant= large, big, sizable
canopy= top, covering, roof
photovoltaic= able to produce electricity from light
panel= board, pane, sheet
micro= very small
turbine= a type of machine through which liquid or gas flows and turns a special wheel with blades in order to produce power

Freiburg Mage Solar Stadium in Germany is the first of a new **wave** of stadiums as power plants, which also includes the Amsterdam Arena and the Kaohsiung Stadium. The latter, **inaugurated** in 2009, has 8,844 photovoltaic panels producing up to 1.14 GWh of electricity annually. This reduces the annual output of carbon dioxide by 660 tons and supplies up to 80 percent of the surrounding area when the stadium is not **in use.** This is **proof** that a stadium can serve its city, and have a **decidedly** positive impact in terms of reduction of CO_2 emissions.

wave= trend, tendency, movement
inaugurate= install, launch, initiate
in use= working, in operation, active
proof= evidence, confirmation, facts
decidedly= definitely, obviously, undoubtedly

G. Sporting arenas have always been **central** to the life and culture of cities. In every **era**, the stadium has **acquired** new value and uses: from **military** fortress to **residential** village, public space to theatre and most recently a **field** for **experimentation** in **advanced** engineering. The stadium of today now **brings together** multiple functions, thus helping cities to create a **sustainable** future.

central= vital, essential, key
era= period, time, age
acquire= get, obtain, gain
military= armed, soldierly, fighting
residential= housing, inhabited, populated
field= ground, arena, pitch
experimentation= research, testing, investigation
advanced= developed, superior, sophisticated
bring together= combine, mix, gather
sustainable= maintainable, supportable, defensible

* amphitheatre: (especially in Greek and Roman architecture) an open circular or oval building with a central space surrounded by tiers of seats for spectators, for the presentation of dramatic or sporting events

TEST 1

READING PASSAGE 3

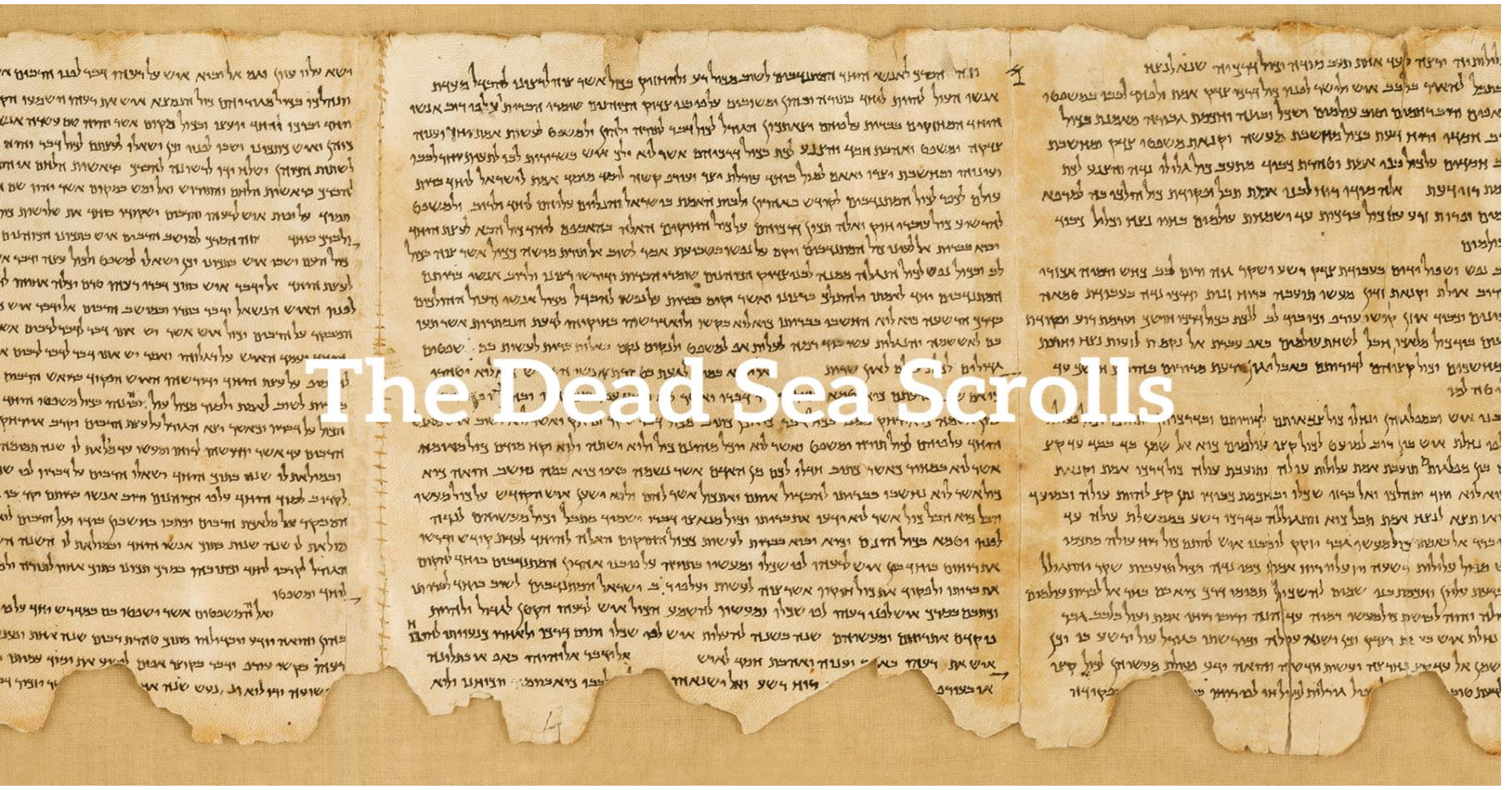
*Anna Keay reviews Charles Spencer's book about the **hunt** for King Charles II during the English Civil War of the seventeenth century*	**hunt**= pursuit, search, chase
Charles Spencer's latest book, *To Catch a King,* tells us the story of the hunt for King Charles II in the six weeks after his **resounding defeat** at the Battle of Worcester in September 1651. And what a story it is. After his father was **executed** by the **Parliamentarians** in 1649, the young Charles II **sacrificed** one of the very **principles** his father had died for and did a **deal** with the Scots, **thereby** accepting Presbyterianism* as the national religion **in return for** being **crowned** King of Scots. His arrival in Edinburgh **prompted** the English Parliamentary **army** to **invade** Scotland in a **pre-emptive strike**. This was followed by a Scottish invasion of England. The two sides finally faced one another at Worcester in the west of England in 1651. After being **comprehensively** defeated on the **meadows** outside the city by the Parliamentarian army, the 21-year-old king found himself the subject of a national	**resounding**= very great **defeat**= loss, setback, reverse # victory **execute**= to kill someone as a legal punishment **Parliamentarian**= a supporter of Parliament in the English Civil War; a Roundhead **sacrifice**= give up, let go, lose **principle**= value, standard, norm **deal**= agreement, arrangement, transaction **thereby**= so, thus, in that way **in return for**= as an exchange for something **crown**= to make someone officially a king or queen of a country **prompt**= encourage, stimulate, provoke **army**= military, defense force, soldiers **invade**= attack, conquer, occupy **pre-emptive strike**= a surprise attack that is launched in order to prevent the enemy from doing it to you **comprehensively**= completely, totally **meadow**= field, grazing land, pasture

manhunt, with a huge **sum** offered for his **capture**. Over the following six weeks he managed, through a series of heart-poundingly close **escapes**, to **evade** the Parliamentarians before seeking **refuge** in France. For the next nine years, the **penniless** and defeated Charles **wandered** around Europe with only a small group of loyal supporters.	**sum**= an amount of money **capture**= arrest, seizure, imprisonment **escape**= running away, getaway, breakout **evade**= avoid, stay away from, steer clear **refuge**= place of safety, protection, sanctuary **penniless**= poor, impoverished, broke **wander**= walk, stroll, roam
Years later, after his **restoration** as king, the 50-year-old Charles II **requested** a meeting with the writer and **diarist** Samuel Pepys. His intention when asking Pepys to **commit his story to paper** was to ensure that this most extraordinary **episode** was never forgotten. Over two three-hour **sittings**, the king **related to** him in great detail his personal **recollections** of the six weeks he had spent as a **fugitive**. As the king and secretary **settled down** (a scene that is surely a gift for a future **scriptwriter**), Charles **commenced** his story: 'After the battle was so absolutely lost as to be beyond hope of recovery, I began to think of the best way of saving myself.	**restoration**= return, re-establishment, reinstatement **request**= ask for, demand, invite **diarist**= writer, biographer, journalist **commit sth to paper**= to write something down **episode**= event, incident, affair **sitting**= meeting, session, appointment **relate**= tell, speak about, narrate **recollection**= memory, recall, reminiscence **fugitive**= a person who is running away or hiding from the police or a dangerous situation **settle down**= relax, calm down, slow down **scriptwriter**= someone who writes stories for movies, television programs, etc **commence**= begin, start, originate
One of the joys of Spencer's book, a result not least of its use of Charles II's own **narrative** as well as those of his supporters, is just how close the reader gets to the action. The day-by-day retelling of the fugitives' **doings** provides **delicious** details: the cutting of the king's long hair with agricultural shears, the use of walnut leaves to **dye** his pale skin, and the day Charles spent lying on a branch of the great oak tree in Boscobel Wood as the Parliamentary soldiers **scoured** the forest floor below. Spencer **draws out** both the humour - such as the **preposterous** refusal of Charles's friend Henry Wilmot to adopt **disguise** on the grounds that it was **beneath his dignity** - and the emotional **tension** when the secret of the king's presence was **cautiously** revealed to his supporters.	**narrative**= description, story, tale **doings**= someone's activities **delicious**= enjoyable, pleasant, appealing **dye**= change the color of, tint, color **scour**= to search a place or thing very carefully in order to try to find something **draw out**= lengthen, make last, prolong **preposterous**= silly, laughable, ridiculous **disguise**= mask, camouflage, concealment **beneath your dignity**= If something is beneath your dignity, you feel that you are too important to do it **tension**= pressure, tightness, stiffness **cautiously**= with care, carefully, watchfully
Charles's adventures after losing the Battle of Worcester hide the uncomfortable truth that **whilst** almost everyone in England had been **appalled** by the **execution** of his father, they had not welcomed the arrival of his son with the Scots army, but had instead firmly **bolted** their doors. This was partly because he rode at the **head** of what looked like a foreign invasion force and partly because, after almost a decade of **civil war**, people were **desperate** to avoid it beginning again. This makes it all the more interesting that Charles II himself loved the story so much ever after. As well as retelling it to anyone who would listen, causing eye rolling among **courtiers**, he **set in train** a series of **initiatives** to **memorialise** it. There was to be a new order of **chivalry**, the Knights of the Royal Oak. A series of enormous oil paintings **depicting** the episode were produced, including a two-metre-wide **canvas** of Boscobel Wood and a set of six similarly enormous paintings of the king **on the run**. In 1660, Charles II **commissioned** the artist John Michael Wright to paint a flying **squadron** of cherubs* carrying an oak tree to the heavens on the ceiling of his **bedchamber**. It is hard to imagine many other kings	**whilst**= while, whereas, although **appalled**= shocked, horrified, disgusted **execution**= the death sentence, killing, putting to death **bolt**= fasten, lock, secure **head**= top, peak, summit **civil war**= a war fought by different groups of people living in the same country **desperate**= determined, eager, in urgent need **courtier**= a companion of a queen, king, or other ruler in their official home **set in train**= to start a process **initiative**= plan, scheme, programme **memorialise**= honor, celebrate, remember **chivalry**= the system of behaviour followed by knights in the medieval period **depict**= portray, illustrate, represent **canvas**= strong, rough cloth used for painting **on the run**= running, fleeing, escaping **commission**= order, assign, appoint **squadron**= a military force consisting of a group of aircraft or ships **bedchamber**= a bedroom

marking the lowest point in their life so enthusiastically, or indeed **pulling off** such an escape in the first place.	**pull off**= to succeed in doing something difficult or unexpected:
Charles Spencer is the perfect person to **pass the story on** to a new generation. His **pacey**, readable **prose steers deftly clear of** modern idioms and elegantly brings to life the details of the great tale. He has **even-handed** sympathy for both the **fugitive** king and the **fierce** republican **regime** that hunted him, and he succeeds in his desire to explore far more of the background of the story than previous books on the subject have done. Indeed, the opening third of the book is about how Charles II found himself at Worcester in the first place, which for some will be reason alone to read *To Catch a King.*	**pass sth on**= hand sth down, retell, continue **pacey**= fast-paced, action-packed **prose**= writing style, text, style **steers clear of**= avoid, omit, reject **deftly**= skillfully, cleverly, smartly **even-handed**= fair, balanced, impartial **fugitive**= escapee, runaway, absconder **fierce**= violent, aggressive, brutal **regime**= government, administration, management
The **tantalising** question left, in the end, is that of what it all meant. Would Charles II have been a different king had these six weeks never happened? The days and nights spent in hiding must have affected him in some way. Did the need to **assume** disguises, to survive on wit and charm alone, to use **trickery** and **subterfuge** to escape from tight corners help form him? This is the one area where the book doesn't quite hit the **mark**. Instead its depiction of Charles II in his final years as an ineffective, pleasure-loving **monarch** doesn't **do justice to** the man (neither is it accurate), or to the complexity of his character. But this one **niggle** aside, *To Catch a King* is an excellent **read**, and those who come to it knowing little of the famous tale will find they have a **treat** in store. •Presbyterianism: part of the reformed Protestant religion •Cherub: an image of angelic children used in paintings	**tantalising**= exciting, alluring, provoking **assume**= use, adopt. acquire **trickery**= dishonesty, fraud, deception **subterfuge**= trick, deception, artifice **mark**= an intended result or an object aimed at **monarch**= ruler, king, queen **do justice to sb/sth**= to treat someone or something in a way that is fair and shows their or its true qualities **niggle**= doubt, worry, concern **read**= the act of reading something **treat**= delight, fun, pleasure

TEST 2

READING PASSAGE 1



In late 1946 or early 1947, three Bedouin teenagers were **tending** their goats and sheep near the **ancient settlement** of Qumran, **located** on the northwest shore of the Dead Sea in what is now known as the West Bank. One of these young **shepherds** **tossed** a rock into an **opening** on the side of a cliff and was surprised to hear a **shattering** sound. He and his **companions** later entered the cave and **stumbled across** a collection of large clay jars, seven of which **contained scrolls** with writing on them. The teenagers took the seven scrolls to a nearby town where they were sold for a small **sum** to a local **antiquities dealer**. **Word of the find spread**, and Bedouins and **archaeologists eventually unearthed** tens of thousands of additional scroll **fragments** from 10 nearby caves; together they **make up** between 800 and 900 manuscripts. It soon became clear that this was one of the greatest archaeological discoveries ever made.

tend= manage, watch, supervise
ancient= outdate, old-fashioned, antiquated
settlement= community, society, village
locate= place, situate, position
shepherd= sheep herder, sheepmen, sheepwomen
toss= throw, pitch, lob
opening= gap, hole, notch
shattering= crushing, smashing, wrecking
companion= friend, colleague acquaintance
stumble across = find, discover, come across
contain= include, surround, comprise
scroll= manuscript, document, copy
sum= a particular amount of money
antiquity= relic, antique, artefact
dealer= trader, seller, wholesaler
spread the word= to communicate a message to a lot of people
archaeologist= someone who studies the buildings, graves, tools, and other objects of people who lived in the past
eventually= finally, ultimately, sooner or later
unearth= uncover, discover, reveal
fragment= piece, portion, part
make up= form, comprise, constitute

The **origin** of the Dead Sea Scrolls, which were written around 2,000 years ago between 150 BCE and 70 CE, is still the subject of **scholarly debate** even today. According to the **prevailing** theory, they are the **work** of a population that **inhabited** the area until Roman **troops** destroyed the settlement around 70 CE. The area was known as Judea at that time, and the people are thought to have belonged to a group called the Essenes, a **devout** Jewish **sect**.	**origin=** used to describe the particular way in which something started to exist **manuscript=** copy, text, document **scholarly**= relating to serious study of a particular subject **debate**= discussion, argument, dispute **prevailing**= current, existing, **inhabit**= occupy, settle, dwell **work**= production, creation **troop**= soldiers or armed forces. **devout**= sincere, honest, earnest **sect**= a group of people with their own particular set of beliefs and practices, especially within or separated from a larger religious group
The majority of the texts on the Dead Sea Scrolls are in Hebrew, with some **fragments** written in an ancient version of its alphabet thought to have **fallen out of** use in the fifth century BCE. But there are other languages as well. Some scrolls are in Aramaic, the language spoken by many **inhabitants** of the region from the sixth century BCE to the **siege** of Jerusalem in 70 CE. In addition, several texts **feature translations** of the Hebrew Bible into Greek.	**fragment**= piece, part **fall out of=** to be used no longer **inhabitant**= resident, occupant, dweller **siege=** blockade, barrier, obstruction **feature**= include, highlight, appear **translation**= interpretation, rendition, change
The Dead Sea Scrolls include fragments from every book of the Old Testament of the Bible except for the Book of Esther. The only entire book of the Hebrew Bible **preserved** among the manuscripts from Qumran is Isaiah; this copy, **dated to** the first century BCE, is considered the earliest **biblical** manuscript still in existence. Along with biblical texts, the scrolls include documents about **sectarian regulations** and religious **writings** that do not appear in the Old Testament.	**preserve**= conserve, maintain, sustain **date to**= establish or ascertain the date of (an object or event). **biblical=** the holy book of the Christian religion **sectarian**= religious, rigid, sectional **regulation**= rule, instruction, guideline **writing**= text, script, inscription
The writing on the Dead Sea Scrolls is mostly in black or **occasionally** red ink, and the scrolls themselves are nearly all made of either parchment (animal skin) or an early form of paper called 'papyrus'. The only **exception** is the scroll numbered 3Q15, which was created out of a combination of copper and tin. Known as the Copper Scroll, this **curious** document features letters **chiselled** onto metal - perhaps, as some have **theorized**, to better **withstand the passage of time** One of the most **intriguing** manuscripts from Qumran, this is a sort of ancient treasure map that lists dozens of gold and silver **caches**. Using an **unconventional** vocabulary and odd spelling, it describes 64 underground hiding places that **supposedly** contain **riches** buried for **safekeeping**. None of these **hoards** have been recovered, possibly because the Romans **pillaged** Judea during the first century CE. According to various **hypotheses**, the treasure belonged to local people, or was **rescued** from the Second Temple before its destruction or never existed to begin with.	**occasionally**= infrequently, uncommonly, seldom **exception**= exclusion, omission, allowance **combination**= arrangement, understanding, permutation **curious**= odd, strange, unusual **chisel**= carve, shape, mold **theorize**= hypothesize, conjecture, imagine **withstand**= endure, survive, resist **the passage of time**= the passing of time **intriguing**= fascinating, interesting, exciting **cache**= supply, accumulation, collection **unconventional**= strange, unusual, odd **supposedly**= allegedly, evidently, apparently **rich=** material, asset, resource **safekeeping**= protection, charge, security **hoard=** pile, store, supply **pillage**= if soldiers pillage a place in a war, they steal a lot of things and do a lot of damage **hypothesis**= theory, premise, suggestion **rescue**= save, free, liberate
Some of the Dead Sea Scrolls have been on interesting journeys. In 1948, a Syrian Orthodox **archbishop** known as Mar Samuel **acquired** four of the original seven scrolls from a Jerusalem shoemaker and part-time **antiquity dealer**, paying less than $100 for them. He then travelled to the United States and	**archbishop**= a priest of the highest rank, who is in charge of all the churches in a particular area **acquire**= get, gain, obtain **antiquity dealer**= a person engaged in the business of selling antiques

unsuccessfully offered them to a number of universities, including Yale. Finally, in 1954, he placed an advertisement in the business newspaper *The Wall Street Journal* - under the category '**Miscellaneous** Items for Sale' - that **read**: 'Biblical Manuscripts dating back to at least 200 B.C. are for sale. This would be an ideal gift to an educational or religious **institution** by an individual or group.' Fortunately, Israeli archaeologist and **statesman** Yigael Yadin **negotiated** their purchase and brought the scrolls back to Jerusalem, where they remain to this day.	**miscellaneous**= various, assorted, diverse **read**= state, say, announce **institution**= organization, establishment, association **statesman=** a political or government leader, especially one who is respected as being wise and fair **negotiate=** discuss, reach a deal, bargain
In 2017, researchers from the University of Haifa restored and **deciphered** one of the last untranslated scrolls. The university's Eshbal Ratson and Jonathan Ben-Dov spent one year **reassembling** the 60 fragments that make up the scroll. Deciphered from a band of coded text on parchment, the **find** provides **insight** into the community of people who wrote it and the 364-day calendar they would have used. The scroll names celebrations that **indicate** shifts in seasons and details two yearly religious events known from another Dead Sea Scroll. Only one more known **scroll** remains untranslated.	**decipher=** to change a message written in a code into ordinary language so that you can read it **reassemble=** reconvene, reunite, congregate **find=** discovery, invention **insight=** vision, understanding **indicate=** show, specify, direct

TEST 2

READING PASSAGE 2

A. It took at least 3,000 years for humans to learn how to **domesticate** the wild tomato and **cultivate** it for food. Now two separate teams in Brazil and China have done it all over again in less than three years. And they have done it better in some ways, as the re-domesticated tomatoes are more **nutritious** than the ones we eat at present.	**domesticate=**tame, control, housetrain **cultivate=** nurture, farm, grow **nutritious=** healthy, healthful, nourishing
This **approach relies on** the **revolutionary** CRISPR **genome** editing technique, in which changes are **deliberately** made to the DNA of a living **cell**, allowing **genetic material** to be added, removed or **altered**. The technique could not only improve existing crops, but could also be used to turn thousands of wild plants into useful and **appealing** foods. In fact, a third team in the US has already begun to do this with a relative of the tomato called the groundcherry.	**approach=** method, technique, tactic **rely on=** depend on, count on, bank on **revolutionary=** groundbreaking, innovative, progressive **genome=** all the genes in one type of living thing → DNA **deliberately=** consciously thoughtfully, purposely **cell**= the smallest part of a living thing that can exist independently **genetic=** relating to genes or genetics **material=** substance, item, object **alter=** modify, change, adjust **appealing**= attractive, tempting, alluring

This **fast-track domestication** could help make the world's food supply healthier and far more **resistant** to diseases, such as the **rust fungus devastating** wheat crops.	**fast-track**= happening or making progress more quickly than is usual **domestication=** housetraining, taming **resistant**= resilient, tolerant, unaffected **rust**= a plant disease that causes reddish-brown spots **fungus=** mushroom, mold **devastate=** destroy, damage, harm
'This could **transform** what we eat,' says Jorg Kudla at the University of Munster in Germany, a member of the Brazilian team. 'There are 50,000 **edible** plants in the world but 90 percent of our energy comes from just 15 crops.'	**edible**= something that is edible can be eaten **transform=** change, alter, convert
'We can now **mimic** the known domestication **course** of major crops like rice, **maize**, **sorghum** or others,' says Caixia Gao of the Chinese Academy of Sciences in Beijing. 'Then we might try to domesticate plants that have never been domesticated.'	**mimic=** imitate, impersonate, take off **maize**= corn **sorghum=** a type of grain that is grown in tropical areas **course**= the often gradual development of something
B. Wild tomatoes, which are **native to** the Andes region in South America, produce pea-sized fruits. Over many generations, peoples such as the Aztecs and Incas transformed the plant by selecting and **breeding** plants with mutations* in their genetic structure, which resulted in **desirable traits** such as larger fruit.	**native to=** refer to plants and animals that grow naturally in a place **breed=** raise, farm, produce **desirable=** wanted, looked-for, worthy **trait=** characteristic, feature, mannerism
But every time a single plant with a mutation is taken from a larger **population** for breeding, much genetic **diversity** is lost. And sometimes the desirable mutations come with less **desirable** traits. For instance, the tomato **strains** grown for supermarkets have lost much of their flavour.	**population**= all the people or animals of a particular type who live in one place **diversity=** variety, assortment, mixture **desirable=** wanted, needed, attractive **strain=** an animal or plant from a particular group whose characteristics are different from others
By comparing the genomes of modern plants to those of their wild relatives, **biologists** have been **working out** what genetic changes **occurred** as plants were domesticated. The teams in Brazil and China have now used this knowledge to **reintroduce** these changes **from scratch** while **maintaining** or even **enhancing** the desirable traits of wild strains.	**biologist=** natural scientist, environmentalist, ecologist **work out**= solve, figure out, understand **occur=** happen, take place, arise **reintroduce=** reestablish, reinstate, bring back **from scratch**= if you start something from scratch, you begin it without using anything that existed or was prepared before **maintain=** keep up, sustain, continue **enhance=** improve, develop, advance
C. Kudla's team made six changes altogether. For instance, they **tripled** the size of fruit by editing a gene called FRUIT WEIGHT, and increased the number of tomatoes per **truss** by editing another called MULTIFLORA.	**triple**= to make something increase three times in size **truss**= the stem that carries the flowers, which turn into tomatoes
While the **historical** domestication of tomatoes reduced levels of the red **pigment** lycopene - thought to have **potential** health benefits - the team in Brazil **managed to boost** it instead. The wild tomato has twice as much **lycopene** as **cultivated** ones; the newly domesticated one has five times as much. 'They are quite tasty,' says Kudla. 'A little bit **strong**. And very **aromatic**.'	**historical**= ancient, antique, old **pigment**= color, coloring, tone **potential**= possible, ability, probable **manage to**= to succeed in doing or dealing with something, especially something difficult **boost**= enhance, increase, improve **lycopene**= a red carotenoid pigment present in tomatoes and many berries and fruits. **cultivate**= nurture, farm, grow **strong**= great, intense, extreme **aromatic**= fragrant, sweet-smelling, perfumed

The team in China re-domesticated several strains of wild tomatoes with desirable traits lost in domesticated tomatoes. In this way they managed to create a strain resistant to a common disease called **bacterial** spot race, which can **devastate yields**. They also created another strain that is more salt **tolerant** - and has higher levels of vitamin C.	**bacterial=** very small living things, some of which cause illness or disease **devastate=** destroy, demolish, ruin **yield**= harvest, crop **tolerant=** to continue existing despite bad or difficult conditions
D. Meanwhile, Joyce Van Eck at the Boyce Thompson Institute in New York state decided to use the same approach to domesticate the groundcherry or goldenberry *(Physalis pruinosa)* for the first time. This fruit looks similar to the closely related Cape gooseberry *(Physa/is peruviana).* Groundcherries are already sold to a **limited extent** in the US but they are hard to produce because the plant has a **sprawling** growth habit and the small fruits fall off the branches when **ripe**. Van Eck's team has edited the plants to increase fruit size, make their growth more **compact** and to stop fruits dropping. 'There's potential for this to be a **commercial** crop,' says Van Eck. But she adds that taking the work further would be expensive because of the need to pay for a **licence** for the CRISPR technology and get **regulatory approval**.	**limited=** incomplete, partial, restricted **extent=** degree, level , amount **sprawling**= extensive, expansive, spreading **ripe=** full-grown, mature **compact=** dense, solid, compressed **commercial=** profitable, money making, viable **licence=** permission, authority, right **regulatory=** relating to the activity of checking whether a business is working according to official rules or laws **approval=** official permission
E. This approach could boost the use of many **obscure** plants, says Jonathan Jones of the Sainsbury Lab in the UK. But it will be hard for new foods to grow so popular with farmers and consumers that they become new **staple** crops, he thinks. The three teams already **have their eye on** other plants that could be '**catapulted** into the **mainstream'**, including foxtail, oat-grass and cowpea. By choosing wild plants that are **drought** or heat tolerant, says Gao, we could create crops that will **thrive** even as the planet warms. But Kudla didn't want to **reveal** which species were in his team's sights, because CRISPR has made the process so easy. 'Any one with the right skills could go to their lab and do this.' *mutations: changes in an organism's genetic structure that can be passed down to later generations	**obscure=** unknown, unseen, strange **staple=** a basic food **have sb's eye on=** to have seen something that you want and intend to get **be catapulted into something=** to suddenly experience a particular state, such as being famous **mainstream**= a common thing **drought=** a long period of dry weather when there is not enough water for plants and animals to live **thrive=** flourish, prosper, succeed **reveal=** disclose, expose, uncover

TEST 2

READING PASSAGE 3

***T**wo scientists consider the **origins** of **discoveries** and other **innovative** behavior*	**origin=** root, background, foundation **discovery=** detection, finding, outcome **innovative=** creative, inventive, pioneering
Scientific discovery is popularly believed to **result from** the **sheer genius** of such **intellectual** stars as **naturalist** Charles Darwin and **theoretical physicist** Albert Einstein. Our view of such **unique contributions** to science often **disregards** the person's **prior** experience and the efforts of their **lesser**-known **predecessors**. **Conventional wisdom** also **places great weight on insight** in **promoting breakthrough** scientific achievements, as if ideas **spontaneously pop** into someone's head - fully formed and **functional.**	**result from**= be caused by, arise from, originate from **sheer**= pure, absolute, complete **genius**= mastermind, brilliance, outstanding ability **intellectual=** intelligent, scholarly, knowledgeable **naturalist=** biologist, botanist, natural scientist **theoretical=** hypothetical, academic, abstract **physicist**= a scientist who has special knowledge and training in physics **unique=** exclusive, exceptional, only one of its kind **contribution**= influence, role, involvement **disregard=** ignore, disrespect, neglect **prior**= previous, preceding, past **lesser-known**= less popular **predecessor**= something that comes before another thing in time **conventional=** usual, normal, typical **wisdom=** understanding, knowledge, sense **place emphasis, importance, etc. on something**= highlight, value, stress **weight**= importance, significance, meaning **insight**= vision, awareness, intuition **promote=** stimulate, foster, encourage **breakthrough=** pivotal, central, important **spontaneously**= impulsively, suddenly, naturally **pop into one's head**= suddenly have an idea **functional=** useful, practical, purposeful

There may be some limited truth to this view. However, we believe that it largely **misrepresents** the real **nature** of scientific discovery, as well as that of creativity and innovation in many other **realms** of human **endeavor**. **Setting aside** such greats as Darwin and Einstein - whose **monumental** contributions are **duly** celebrated - we suggest that innovation is more a process of trial and error, where two steps forward may sometimes come with one step back, as well as one or more steps to the right or left. This evolutionary view of human innovation **undermines** the **notion** of creative genius and recognizes the **cumulative** nature of scientific progress.	**misrepresent=** not tell the truth, pretend, lie **nature=** quality, features, character **realm=** field, area, domain **endeavor=** attempt, effort, try **set aside=** to ignore or not think about a particular fact or situation while considering a matter **monumental=** colossal, massive, gigantic. **duly**= accordingly, suitably, appropriately **undermine=** weaken, destabilize, threaten **notion=** belief, concept, perception **cumulative=** aggregate, accumulative, growing
Consider one **unheralded** scientist: John Nicholson, a mathematical physicist working in the 1910s who **postulated** the existence of **'proto**-elements' in outer space. By combining different numbers of weights of these proto-elements' **atoms**, Nicholson could recover the weights of all the elements in the then-known **periodic table**. These successes are all the more(even more) **noteworthy** given the fact that Nicholson was wrong about the presence of proto-elements: they do not actually exist. Yet, amid his often **fanciful** theories and **wild speculations**, Nicholson also proposed a novel theory about the structure of atoms. Niels Bohr, the Nobel prize-winning **father of** modern atomic theory, jumped off from this interesting idea to **conceive** his now-famous model of the atom.	**unheralded=** not known about or recognized as good **postulate=** hypothesize, assume,theorize **proto**= first, especially from which other similar things develop; original **atom=** particle, subdivision, element **periodic table=** a list of the symbols of all the chemical elements arranged in rows and columns down a page **noteworthy=** notable, striking, remarkable **fanciful=** imaginary, make-believe, fictional **wild speculation=** something that you say that is not based on facts and is probably wrong **the father of**= someone who began, or first made something important **conceive**= create, invent, form
What are we **to make of** this story? One might simply conclude that science is a **collective** and cumulative **enterprise**. That may be true, but there may be a deeper insight to be **gleaned**. We propose that science is **constantly** evolving, much as species of animals do. In biological systems, **organisms** may display new characteristics that result from random genetic mutations. In the same way, random, **arbitrary** or accidental mutations of ideas may help **pave the way for advances** in	**make something of something/someone**= to have an impression or an understanding about something **collective=** cooperative, communal, joint **enterprise=** a large project **glean**= pick up, gather, collect **constantly**= continually, continuously, regularly **organism**= creature, being, living things **arbitrary=** random, chance, haphazard **pave the way for=** to make it possible for someone to do something or for something to happen **advance**= development, growth, expansion

science. If mutations prove beneficial, then the animal or the scientific theory will continue to **thrive** and perhaps **reproduce**. Support for this **evolutionary** view of behavioral innovation comes from many **domains**. Consider one example of an **influential** innovation in US horseracing. The so-called 'acey-deucy' **stirrup** placement, in which the rider's foot in his left stirrup is placed as much as 25 centimeters lower than the right, is believed to **confer** important speed advantages when turning on oval tracks. It was developed by a relatively unknown jockey named Jackie Westrope. Had Westrope **conducted methodical investigations** or examined **extensive** film records in a **shrewd** plan to **outrun** his rivals? Had he **foreseen** the speed advantage that would be conferred by riding acey-deucy? No. He suffered a leg injury, which left him unable to fully bend his left knee. His **modification** just happened to **coincide** with enhanced left-hand turning performance. This led to the rapid and widespread **adoption** of riding acey-deucy by many riders, a racing style that continues in today's **thoroughbred** racing.	**thrive**= flourish, prosper, grow **reproduce=** to produce a copy of something **evolutionary**= involving a gradual process of change and development **domain=** area, field **influential**= powerful, important, significant **stirrup=** one of a pair of pieces that hang from the side of a horse's saddle, used for resting your foot when you are riding **confer=** give, provide, grant **conduct**= do, perform, carry out **methodical=** logical, systematic **investigation**= study, examination, exploration **extensive=** wide, large-scale, wide-ranging **shrewd=** wise, cunning, clever **outrun**= run faster than, beat, overtake **foresee**= predict, forecast, anticipate **modification**= alteration, adjustment, change **coincide=** happen together overlap, match **adoption**= accepting or starting to use something new **thoroughbred**= (animals) with parents that are of the same breed and have good qualities
Plenty of other stories show that fresh advances can arise from error, **misadventure**, and also pure **serendipity** - a happy accident. For example, in the early 1970s, two employees of the company 3M each had a problem: Spencer Silver had a product - a glue which was only slightly sticky - and no use for it, while his colleague Art Fry was trying to figure out how to **affix** temporary bookmarks in his hymn book without damaging its pages. The solution to both these problems was the invention of the brilliantly simple yet **phenomenally** successful Post-It note. Such examples **give lie to** the claim that **ingenious**, **designing** minds are responsible for human creativity and invention. Far more **banal** and **mechanical** forces may be at work; forces that are **fundamentally** connected to the laws of science.	**misadventure**= accident, misfortune, mishap **serendipity**= luck, chance, fate **affix=** stick, fasten, attach **phenomenally=** remarkably, unusually, oddly **give the lie to**= to prove that something is not true **ingenious=** clever, resourceful, inventive **designing**= used to describe someone who tries to get what they want, usually dishonestly **banal**= boring, ordinary, not original **mechanical=** without thinking about what you are doing, esp. because you do it often-repetitive **fundamentally**= basically, essentially, primarily

The notions of insight, creativity and genius are often **invoked**, but they remain **vague** and of doubtful scientific utility, especially when one considers the diverse and enduring contributions of individuals such as Plato, Leonardo da Vinci, Shakespeare, Beethoven, Galileo, Newton, Kepler, Curie, Pasteur and Edison. These notions **merely** label rather than explain the evolution of human innovations. We need another approach, and there is a promising candidate.	**invoke=** mention, refer, quote **vague=** unclear, abstracted, dreamy **merely=** simply, just, only
The Law of Effect was **advanced** by **psychologist** Edward Thorndike in 1898, some 40 years after Charles Darwin published his **groundbreaking** work on biological evolution, *On the Origin of Species.* This simple law **holds** that organisms tend to repeat successful behaviors and to **refrain** from performing unsuccessful ones. Just like Darwin's Law of Natural Selection, the Law of Effect involves an entirely mechanical process of variation and selection, without any end **objective** in sight.	**advance**= to suggest an idea or theory **psychologist** = someone who is trained in psychology **groundbreaking=** revolutionary, innovative, advanced **hold=** to state that something is true **refrain**= avoid doing, cease, hold back **refrain=** desist, abstain, renounce **objective**= purpose, goal, intention
Of course, the origin of human innovation demands much further study. In **particular**, the **provenance** of the raw material on which the Law of Effect **operates** is not as clearly known as that of the genetic mutations on which the Law of Natural Selection operates. The generation of novel ideas and behaviors may not be entirely random, but **constrained** by prior successes and failures - of the current individual (such as Bohr) or of predecessors (such as Nicholson).	**particular**= specific, precise, exact **provenance**= origin, background, birth place **operate=** work, conduct, carry out **constrain**= restrain, restrict, control
The time seems right for **abandoning** the **naive notions** of intelligent design and genius, and for scientifically exploring the true origins of creative behavior.	**abandon=** end, leaving, cancel **naive=** simple, childlike, innocent **notion=** idea, view, concept

TEST 3

READING PASSAGE 1

The **extinct** thylacine, also known as the Tasmanian tiger, was a marsupial* that **bore a superficial resemblance** to a dog. Its most **distinguishing feature** was the 13- 19 dark brown **stripes** over its back, beginning at the **rear** of the body and **extending** onto the tail. The thylacine's average nose to-tail length for adult males was 162.6 cm, compared to 153.7 cm for females.	**extinct**= died out, wiped out, nonexistent **superficial**= external, exterior, apparent **bare a resemblance**= to look a lot like someone else **distinguishing**= unique, distinctive, differentiating **feature**= characteristic, trait, attribute **stripe=** line, strip, bar **rear**= at the back of something **extend**= continue, reach, go on
The thylacine appeared to **occupy** most types of **terrain** except **dense rainforest**, with open **eucalyptus** forest thought to be its **prime** habitat. In terms of feeding, it was **exclusively carnivorous**, and its stomach was **muscular** with an ability to **distend** so that it could eat large amounts of food at one time,	**occupy**= inhabit, live in, dwell in **terrain**= land, topography, ground **dense**= thick, concentrated, compact **rainforest**= a tropical forest with tall trees that are very close together, growing in an area where it rains a lot **eucalyptus**= a tall tree that produces an oil with a strong smell, used in medicines **prime**= main, primary **exclusively**= solely, wholly, uniquely **carnivorous**= meat-eating, flesh-eating **muscular**= strong, powerful **distend**= swell up, expand, enlarge

probably an **adaptation** to **compensate** for long periods when hunting was unsuccessful and food **scarce**. The thylacine was not a fast runner and probably caught its **prey** by **exhausting** it during a long **pursuit**. During long-distance chases, thylacines were likely to have relied more on scent than any other sense. They **emerged** to hunt during the evening, night and early morning and tended to **retreat** to the hills and forest for shelter during the day. Despite the common name ' tiger', the thylacine had a shy, nervous **temperament**. Although mainly **nocturnal**, it was sighted moving during the day and some individuals were even recorded **basking** in the sun.	**adaptation**= adjustment, modification, change **compensate**= balance, pay off, offset **scarce**= rare, limited, inadequate **prey**= an animal that is hunted and killed for food by another animal: **exhaust**= tire, drain, weaken **pursuit**= chase, hunt, track down **emerge**= appear, come out, come into view **retreat**= go back, retire, hide **temperament**= nature, character, personality **nocturnal**= active at night **bask**= to lie or sit enjoying the warmth especially of the sun
The thylacine had an extended breeding season from winter to spring, with **indications** that some breeding took place **throughout** the year. The thylacine, like all marsupials, was tiny and hairless when born. Newborns crawled into the **pouch** on the belly of their mother, and attached themselves to one of the four **teats**, remaining there for up to three months. When old enough to leave the pouch, the young stayed in a **lair** such as a deep rocky cave, well-hidden nest or **hollow** log, **whilst** the mother hunted.	**indication**= hint, sign, suggestion **throughout**= all over, during, all the way through **pouch**= a pocket of skin on the stomach which marsupials such as kangaroos use for carrying their babies **teat**= one of the small parts on a female animal's body that her babies suck milk from **lair**= the place where a wild animal hides and sleeps **hollow**= empty, unfilled, unoccupied **whilst**= at the same time as, concurrently, while
Approximately 4,000 years ago, the thylacine was **widespread** throughout New Guinea and most of **mainland** Australia, as well as the island of Tasmania. The most recent, **well-dated occurrence** of a thylacine on the mainland is a carbon-dated **fossil** from Murray Cave in Western Australia, which is around 3,100 years old. Its **extinction coincided** closely with the arrival of wild dogs called dingoes in Australia and a similar **predator** in New Guinea. Dingoes never reached Tasmania, and most scientists see this as the main reason for the thylacine's survival there.	**approximately**= about, around, roughly **widespread**= common, prevalent, general **mainland**= landmass, continent, interior **well-dated**= able to precisely guess the age **occurrence**= existence, incidence **fossil**= the shape of a bone, a shell, or a plant or animal that has been preserved in rock for a very long period **extinction**= death, disappearance # survival **coincide**= concur, happen together, overlap **predator**= killer, slayer, hunter
The **dramatic** decline of the thylacine in Tasmania, which began in the 1830s and continued for a century, is generally **attributed to** the **relentless** efforts of sheep farmers and bounty hunters** with shotguns. While this determined campaign **undoubtedly** played a large part, it is likely that various other factors also contributed to the decline and **eventual** extinction of the species. These include competition with wild dogs **introduced** by European **settlers**, loss of habitat along with the disappearance of prey species, and a distemper-like disease which may also have affected the thylacine.	**dramatic**= drastic, significant **attribute to**= to say or think that something is the result or work of something or someone else **relentless**= persistent, unyielding, harsh **undoubtedly**= certainly, definitely, unquestionably **eventual**= ultimate, final, concluding **introduce=** to put something into use, operation, or a place for the first time **settler**= colonizer, immigrant, incomer
There was only one successful attempt to breed a thylacine in **captivity**, at Melbourne Zoo in 1899. This was despite the large numbers that went through some zoos, **particularly** London Zoo and Tasmania's Hobart Zoo. The famous **naturalist** John Gould **foresaw** the thylacine's **demise** when he published his *Mammals of Australia* between 1848 and 1863, writing, 'The numbers of	**captivity**= enclosure, detention, confinement **particularly**= especially, specifically, outstandingly **naturalist**= biologist, zoologist, environmentalist **foresee=** predict, forecast, anticipate **demise**= death, loss, decease

this **singular** animal will **speedily diminish**, **extermination** will have its full **sway**, and it will then, like the wolf of England and Scotland, be recorded as an animal of the past.' However, there seems to have been little public pressure to **preserve** the thylacine, nor was much concern expressed by scientists at the decline of this species in the decades that followed. A **notable exception** was T.T. Flynn, Professor of Biology at the University of Tasmania. In 1914, he was **sufficiently** concerned about the **scarcity** of the thylacine to suggest that some should be **captured** and placed on a small island. But it was not until 1929, with the species on the very **edge** of extinction, that Tasmania's Animals and Birds Protection Board passed a **motion** protecting thylacines only for the month of December, which was thought to be their **prime breeding** season. The last known wild thylacine to be killed was shot by a farmer in the north-east of Tasmania in 1930, leaving just **captive specimens**. Official protection of the species by the Tasmanian government was introduced in July 1936, 59 days before the last known individual died in Hobart Zoo on 7th September, 1936. There have been **numerous expeditions** and searches for the thylacine over the years, none of which has produced **definitive** evidence that thylacines still exist. The species was **declared** extinct by the Tasmanian government in 1986. *marsupial: a mammal, such as a kangaroo, whose young are born incompletely developed and are typically carried and suckled in a pouch on the mother's belly **bounty hunters: people who are paid a reward for killing a wild animal	**singular**= unique, outstanding, particular **speedily**= quickly, rapidly, immediately **diminish**= reduce, weaken, fade **extermination**= extinction, termination # preservation **sway**= power, control, influence **preserve**= maintain, protect, conserve **notable**= distinguished, prominent, noteworthy **exception**= exclusion, omission, exemption **sufficiently**= adequately, satisfactorily, appropriately **scarcity**= shortage, lack, insufficiency **capture**= take, seize, catch **edge**= brink, verge, threshold **motion**= a formal suggestion made, discussed, and voted on at a meeting **prime**= best, superior **breeding**= the process in which animals have sex and produce young animals **captive**= caged, imprisoned, in prison **specimen**= example, case, sample **numerous**= many, plentiful, abundant **expedition**= trip, voyage, excursion, **definitive**= conclusive, ultimate, absolute **declare**= state, announce, pronounce

TEST 3
READING PASSAGE 2

A. **P**alm oil is an **edible** oil **derived** from the fruit of the African oil palm tree, and is currently the most **consumed** vegetable oil in the world. It's almost **certainly** in the soap we wash with in the morning, the sandwich we have for lunch, and the biscuits we **snack** on during the day. Why is palm oil so attractive for **manufacturers**? **Primarily** because its **unique properties**- such as remaining **solid** at room temperature - make it an **ideal** ingredient for long-term **preservation**, allowing many **packaged** foods on supermarket shelves to have 'best before' dates of months, even years, into the future.	**edible**= something that is edible can be eaten **derive**= get, obtain, receive **consume=** use, utilize, eat **certainly**= surely, absolutely, definitely **snack**= to eat small amounts of food between main meals **manufacturer**= producer, industrialist, company **primarily**= mainly, essentially, for the most part **unique**= rare, exclusive, exceptional **property=** quality, characteristic **solid**= hard, firm **ideal**= perfect, fitting, suitable **preservation**= maintenance, conservation, continuation **package**= pack, wrap, bundle
B. Many farmers have **seized** the opportunity to maximise the planting of oil palm trees. Between 1990 and 2012, the global land area **devoted** to growing oil palm trees grew from 6 to 17 million hectares, now **accounting for** around ten percent of total cropland in the entire world. From a **mere** two million tonnes of palm oil being produced **annually globally** 50 years ago, there are now around 60	**seize**= capture, grab, grasp **devote**= dedicate, give, offer **account for**= to form part of a total **mere**= used to emphasize how small or unimportant something or someone is **annually**= yearly, once a year, every twelve months **globally**= internationally, worldwide, universally

million tonnes produced every single year, a figure looking likely to double or even triple by the middle of the century.	
C. However, there are **multiple** reasons why **conservationists cite** the rapid **spread** of oil palm **plantations** as a major concern. There are **countless** news stories of **deforestation**, **habitat destruction** and **dwindling** species populations, all as a direct result of land clearing to establish oil palm tree **monoculture** on an industrial scale, particularly in Malaysia and Indonesia. **Endangered** species - most famously the Sumatran **orangutan**, but also rhinos, elephants, tigers, and numerous other **fauna** - have suffered from the unstoppable spread of oil palm plantations.	**multiple**= many, numerous, various **conservationist**= someone who works to protect animals, plants etc or to protect old buildings **cite**= quote, name, mention **spread**= expansion, development, increase **plantation**= farm, agricultural estate, cultivated area **countless**= uncountable, innumerable, immeasurable **deforestation**= the cutting or burning down of all the trees in an area **habitat**= home, environment, locale **destruction**= ruin, damage, devastation **dwindling**= declining, deteriorating, falling **monoculture**= the practice of growing only one crop or keeping only one type of animal on an area of farm land **endangered**= rare, threatened, vulnerable **orangutan**= a large ape with long arms and long orange-brown hair **fauna**= wildlife, creature, animal
D. 'Palm oil is surely one of the greatest threats to global **biodiversity**,' **declares** Dr Farnon Ellwood of the University of the West of England, Bristol. 'Palm oil is replacing **rainforest**, and rainforest is where all the species are. That's a problem.' This has led to some **radical** questions among **environmentalists**, such as whether consumers should try to **boycott** palm oil **entirely**. Meanwhile Bhavani Shankar, Professor at London's School of Oriental and African Studies, argues, 'It's easy to say that palm oil is the enemy and we should be against it. It makes for a more **dramatic** story, and it's very **intuitive**. But given the **complexity** of the argument, I think a much more **nuanced** story is closer to the truth.'	**biodiversity**= the variety of plants and animals in a particular place **declare**= state, announce, affirm **rainforest**= a tropical forest with tall trees that are very close together, growing in an area where it rains a lot **radical**= fundamental, essential, profound **environmentalist**= someone who is concerned about protecting the environment **boycott**= refuse, avoid, reject **entirely**= completely, totally, absolutely **dramatic**= spectacular, extraordinary, remarkable **intuitive**= instinctive, spontaneous, impulsive **complexity**= difficulty, intricacy, sophistication **nuanced**= made slightly different in appearance, meaning, sound, etc.
E. One response to the **boycott** movement has been the argument for the **vital** role palm oil plays in lifting many millions of people in the developing world out of **poverty**. Is it desirable to have palm oil boycotted, replaced, **eliminated** from the global **supply chain**, given how many low-income people in developing countries depend on it for their **livelihoods**? How best to **strike** a **utilitarian** balance between these **competing** factors has become a serious **bone of contention**.	**vital**= fundamental, imperative, crucial **poverty**= the condition of being extremely poor **eliminate**= remove, eradicate, abolish **supply chain**= the system of getting a product from the place where it is made to the person who buys it **livelihood=** living, income, source of revenue **strike**= hit, reach, achieve **utilitarian**= useful, practical, down-to-earth **competing**= opposing, challenging, rival **a bone of contention=** something that two or more people argue about strongly over a long period of time
F. Even the deforestation argument isn't as **straightforward** as it seems. Oil palm plantations produce at least four and **potentially** up to ten times more oil per hectare than soybean, rapeseed, sunflower or other competing oils. That **immensely** high **yield** - which is **predominantly** what makes it so profitable - is potentially also an **ecological** benefit. If ten times more palm oil can be produced from a **patch** of land than any competing oil, then ten times more land would need to be cleared in order to produce the same **volume** of oil from that **competitor**.	**straightforward**= upfront, uncomplicated, direct **potentially**= possibly, theoretically, hypothetically **immensely**= hugely, enormously, massively **yield=** production, harvest **predominantly**= mainly, principally, primarily **ecological**= environmental, biological, natural **patch**= area, spot, piece **volume**= size, capacity, quantity **competitor**= rival, contestant, opponent
As for the question of carbon **emissions**, the issue really depends on what oil palm trees are replacing. Crops vary in	**emission**= release, discharge, emanation

the degree to which they **sequester** carbon - in other words, the amount of carbon they capture from the atmosphere and store within the plant. The more carbon a plant sequesters, the more it reduces the effect of climate change. As Shankar explains: ' [Palm oil production] actually sequesters more carbon in some ways than other **alternatives**. [...] Of course, if you're cutting down **virgin forest** it's terrible - that's what's happening in Indonesia and Malaysia; it's been allowed to get **out of hand**. But if it's replacing rice, for example, it might actually sequester more carbon.'	**sequester**= to separate and store a harmful substance **alternative**= replacement, substitute, another possibility **virgin forest**= A virgin forest or area of land has not yet been cultivated or used by people **out of hand=** uncontrollable, out of control, unmanageable
G. The industry is now **regulated** by a group called the Roundtable on Sustainable Palm Oil (RSPO), consisting of palm growers, retailers, product manufacturers, and other interested **parties**. Over the past decade or so, an **agreement** has **gradually** been reached regarding standards that producers of palm oil have to meet in order for their product to be regarded as officially ' **sustainable'**. The RSPO **insists** upon no virgin forest clearing, **transparency** and regular **assessment** of carbon stocks, among other **criteria**. Only once these requirements are fully satisfied is the oil allowed to be sold as **certified** sustainable palm oil (CSPO). Recent figures show that the RSPO now certifies around 12 million tonnes of palm oil annually, **equivalent** to **roughly** 21 percent of the world's total palm oil production.	**regulate**= control, adjust, standardize **party**= participant, organization, contributor **agreement**= contract, settlement, deal **gradually**= progressively, steadily, regularly **sustainable**= maintainable, supportable, viable **insist**= require, demand, enforce **transparency**= honesty, without secret **assessment**= valuation, calculation, judgement **criterion**= standard, principle, condition **certified=** qualified, licensed, official **equivalent**= corresponding, comparable, equal **roughly**= approximately, about, around
H. There is even hope that oil palm plantations might not need to be such **sterile** monocultures, or ' green deserts', as Ellwood describes them. New research at Ellwood's lab **hints** at one plant which might make all the difference. The **bird's nest fern** *(Asplenium nidus)* grows on trees in an **epiphytic fashion** (meaning it's dependent on the tree only for support, not for **nutrients**), and is native to many tropical regions, where as a **keystone** species it performs a vital ecological role. Ellwood believes that reintroducing the bird's nest fern into oil palm plantations could potentially allow these areas to recover their **biodiversity**, providing a home for all **manner** of species, from **fungi** and bacteria, to **invertebrates** such as insects, **amphibians**, **reptiles** and even **mammals**.	**sterile**= lacking diversity **hint**= suggest, refer to, mention **bird's nest fern=** a green plant with long stems, leaves like feathers, and no flowers **epiphytic**= relating to a plant that grows on another plant but does not feed from it **fashion**= method, technique **nutrient**= a chemical or food that provides what is needed for plants or animals to live and grow **keystone**= foundation, base, cornerstone **biodiversity=** the number and types of plants and animals that exist in a particular area **manner=** type, kind, sort **fungi**= mushroom, molds, toadstool **invertebrate**= a living creature that does not have a backbone **amphibian**= an animal such as a frog that can live both on land and in water **reptile**= a type of animal whose body temperature changes according to the temperature around it **mammal**= a type of animal that drinks milk from its mother's body when it is young.

TEST 3

READING PASSAGE 3

***K**atharine L. Shester reviews a book by Jason Barr about the development of New York City* In *Building the Skyline,* Jason Barr takes the reader through a **detailed** history of New York City. The book combines geology, history, economics, and a lot of data to explain why business **clusters** developed where they did and how the early decisions of workers and firms shaped the **skyline** we see today. *Building the Skyline* is organized into two **distinct** parts. The first is **primarily** historical and addresses New York's settlement and growth from 1609 to 1900; the second deals primarily with the 20th century and is a **compilation** of chapters commenting on different aspects of New York's **urban** development. The **tone** and organization of the book change **somewhat** between the first and second parts, as the **latter** chapters **incorporate** aspects of Barr's related research **papers.**	**detailed**= thorough, comprehensive, meticulous **cluster**= bunch, group, collection **skyline**= horizon, distance, prospect **distinct=** separate, different, distinctive **primarily**= chiefly, mainly, principally **compilation**= collection, gathering, assembling **urban**= city, metropolitan, rural **tone**= style, spirit **somewhat**= slightly, fairly, to some extent **latter=** second, later, last **incorporate**= merge, include, integrate **paper**= document, record, manuscript
Barr begins chapter one by taking the reader on a 'helicopter time-machine' ride - giving a **fascinating account** of how the New York **landscape** in 1609 might have looked from the sky. He then moves on to a **subterranean** walking tour of the city,	**fascinating**= captivating, charming, intriguing **account=** description, explanation, justification **landscape**= scenery, setting, background **subterranean**= under the ground

indicating the location of rock and water below the **subsoil**, before taking the reader back to the surface. His love of the city comes through as he describes **various** fun facts about the location of the New York **residence** of early 19th-century vice-president Aaron Burr as well as a number of **legends** about the city.	**subsoil**= the layer of soil that is under the surface level **various**= numerous, diverse, assorted **residence**= a home **legend=** fairytale, folklore, folk tale
Chapters two and three take the reader up to the Civil War (1861- 1865), with chapter two focusing on the early development of land and the **implementation** of a **grid** system in 1811. Chapter three focuses on land use before the Civil War. Both chapters are **informative** and well-researched and **set the stage for** the economic analysis that comes later in the book. I would have liked Barr to expand upon his claim that existing **tenements*** prevented **skyscrapers** in certain neighborhoods because 'likely no skyscraper developer was interested in performing the necessary "**slum clearance**"'. Later in the book, Barr makes the claim that the depth of bedrock** was not a limiting factor for developers, as **foundation** costs were a small **fraction** of the cost of development. At first **glance**, it is not obvious why slum clearance would be limiting, while more expensive foundations would not.	**implementation=** installation employment, putting into practice **grid**= a system of wires through which electricity is connected to different power stations **informative**= providing a lot of useful information **set the stage for**= to make it possible for something else to happen **tenement**= a large building divided into apartments, usually in a poor area of a city **skyscraper**= tower, multistory building, high-rise building **slum**= a house or an area of a city that is in very bad condition, where very poor people live **clearance**= removal, erasure **foundation**= groundwork, base, ground **fraction**= portion, segment, part **glance**= look, glimpse,
Chapter four focuses on **immigration** and the location of neighborhoods and tenements in the late 19th century. Barr identifies four primary **immigrant enclaves** and analyzes their locations in terms of the **amenities** available in the area. Most of these enclaves were located on the least **valuable** land, between the industries located on the **waterfront** and the **wealthy** neighborhoods **bordering** Central Park.	**immigration**= migration, arrival, entry **immigrant**= settler, migrant, refugee **enclave**= area, territory, community **amenity**= facility, convenience, comfort **valuable**= appreciated, precious, valued **waterfront=** seaside, waterside, beachfront **wealthy**= rich, affluent, prosperous **border=** be next to, run alongside, be adjacent to
Part two of the book begins with a discussion of the economics of skyscraper height. In chapter five, Barr **distinguishes** between **engineering** height, economic height, and developer height - where engineering height is the tallest building that can be safely made at a given time, economic height is the height that is most **efficient** from society's point of view, and developer height is the actual height chosen by the developer, who is **attempting** to **maximize return on investment.** Chapter five also has an interesting discussion of the technological **advances** that led to the **construction** of skyscrapers. For example, the introduction of iron and steel **skeletal** frames made thick, **load-bearing** walls unnecessary, expanding the usable **square footage** of buildings and increasing the use of windows and **availability** of natural light. Chapter six then presents data on building height **throughout** the 20th century and uses **regression analysis** to 'predict' building construction. While less **technical** than the research paper on which the chapter is based, it is **probably** more technical than would be preferred by a general **audience**.	**distinguish**= analyze, differentiate, discriminate **engineering**= the work involved in designing and building roads, bridges, machines etc **efficient**= effective, proficient, economical **attempt**= try, make an effort, endeavor **maximize**= boost, increase, expand **return on investment** = the profit from an activity compared with the amount invested in it **advance**= development, improvement, breakthrough **construction**= creation, manufacture, composition **skeletal**= of or like a frame of bones **load-bearing**= supporting the weight of the building above it **square footage**= an area measured in feet **availability**= readiness, obtainability, abundance **throughout**= during, the whole time **regression analysis**= a method in statistics that compares the way two or more related sets of numbers have changed over a particular period **technical**= practical, mechanical, methodological **probably**= perhaps, maybe, possibly **audience=** viewers, watchers, spectators

Chapter seven **tackles** the 'bedrock **myth'**, the **assumption** that the **absence** of bedrock close to the surface between Downtown and Midtown New York is the reason for skyscrapers not being built between the two urban centers. Rather, Barr argues that while deeper **bedrock** does increase foundation costs, these costs were neither **prohibitively** high nor were they large compared to the overall cost of building a skyscraper. What I enjoyed the most about this chapter was Barr's discussion of how foundations are actually built. He describes the use of **caissons**, which enable workers to **dig** down for considerable distances, often below the **water table**, until they reach bedrock. Barr's thorough technological history discusses not only how caissons work, but also the dangers involved. While this chapter **references empirical** research papers, it is a **relatively** easy read.	**tackle**= solve, stop, confront **myth=** falsehood, untruth, fiction **assumption**= supposition, hypothesis, statement **absence**= lack, nonexistence, deficiency **bedrock**= foundation, base **prohibitively**= excessively, exorbitantly, expensively **caisson**= a structure that goes under water or underground and keeps water out, used in building **dig**= to form a hole by moving soil **water table**= the level below the surface of the ground at which you start to find water **reference=** link to, quote **empirical**= experiential, observed, practical **relatively**= comparatively, moderately, fairly
Chapters eight and nine focus on the birth of Midtown and the building **boom** of the 1920s. Chapter eight contains **lengthy** discussions of urban economic theory that may serve as a distraction to readers primarily interested in New York. However, they would be well-suited for **undergraduates** learning about the economics of cities. In the next chapter, Barr considers two of the primary explanations for the building boom of the 1920s -the first being **exuberance**, and the second being financing. He uses data to assess the **viability** of these two explanations and finds that supply and demand factors explain much of the development of the 1920s; though it enabled the boom, cheap **credit** was not, he argues, the primary cause.	**boom=** explosion, escalation, surge **lengthy**= long, extensive, long-lasting **undergraduate**= a student at college or university, who is working for their first-degree **exuberance**= lavishness **viability**= feasibility, practicality, capability **credit** =a method of paying for goods or services at a later time
In the final chapter (chapter 10), Barr discusses another of his **empirical** papers that **estimates** Manhattan land values from the mid-19th century to the present day. The data work that went into these estimations is particularly **impressive**. Toward the end of the chapter, Barr assesses 'whether skyscrapers are a cause or an effect of high land values. He finds that changes in land values **predict** future building height, but the **reverse** is not true. The book ends with an **epilogue**, in which Barr discusses the impact of climate change on the city and makes policy **suggestions** for New York going **forward**.	**empirical**= using experience **estimate**= assess, guess, value **impressive**= striking, remarkable, notable **predict**= forecast, foresee, expect **reverse**= opposite, contrary, opposite **epilogue**= a speech or piece of text that is added to the end of a play or book **suggestion**= hint, proposal, implication **forward**= ahead, further, progress

TEST 4

READING PASSAGE 1

How Madagascar's bats are helping to save the **rainforest** There are few places in the world where relations between agriculture and **conservation** are more **strained**. Madagascar's forests are being **converted** to agricultural land at a rate of one percent every year. Much of this **destruction** is **fuelled** by the **cultivation** of the country's main **staple** crop: rice. And a **key** reason for this destruction is that insect **pests** are destroying **vast** quantities of what is grown by local **subsistence** farmers, leading them to clear forest to create new **paddy** fields. The result is **devastating** habitat and **biodiversity** loss on the island, but not all species are suffering. In fact, some of the island's **insectivorous** bats are currently **thriving** and this has important **implications** for farmers and **conservationists** alike.	**rainforest**= a tropical forest with tall trees that are very close together, growing in an area where it rains a lot **conservation=** protection, saving, preservation **strained**= stressed, tense, anxious **convert**= change, switch, move **destruction**= ruin, damage, demolition **fuel**= stimulate, energize, promote **cultivation**= farming, agriculture, gardening **staple**= basic, core, prime **key**= crucial, significant, important **pest=** insect, bug, vermin **vast**= huge, massive, enormous **subsistence**= existence, maintenance, sustenance **paddy**= a field in which rice is grown in water **devastating**= overwhelming, shocking, upsetting **biodiversity**= the variety of plants and animals in a particular place **insectivore** = a creature that eats insects for food **thriving**= blooming, increasing **implication**= suggestion , insinuation, consequence **conservationist**= ecologist, environmentalist, preservationist
Enter University of Cambridge zoologist Ricardo Rocha. He's **passionate** about conservation, and bats. More specifically, he's interested in how bats are responding to human activity and **deforestation** in particular. Rocha's new study shows that several species of bats are giving Madagascar's rice farmers a vital pest control service by **feasting** on **plagues of** insects. And this, he	**passionate**= obsessive, enthusiastic **deforestation**= the cutting or burning down of all the trees in an area **feast**= eat, devour, indulge **a plague of**= a large number of things that are unpleasant or likely to cause damage

believes, can **ease** the financial pressure on farmers to turn forest into fields.	**ease**= relieve, reduce, lessen
Bats **comprise** roughly one-fifth of all **mammal** species in Madagascar and thirty-six recorded bat species are **native** to the island, making it one of the most important regions for conservation of this animal group anywhere in the world.	**comprise**= to be the parts of something; to make up something **mammal**= any animal of which the female feeds her young on milk from her own body **native=** local, indigenous, domestic
Co-leading an international team of scientists, Rocha found that several species of **indigenous** bats are **taking advantage of** habitat **modification** to hunt insects **swarming** above the country's rice fields. They include the Malagasy mouse-eared bat, Major's long-fingered bat, the Malagasy white-bellied free-tailed bat, and Peters' wrinkle-lipped bat.	**indigenous**= local, innate, natural **take advantage of=** make the most of, exploit, make use of **modification**= alteration, change, adjustment **swarm**= crowd, mass, flock
'These winner species are providing a **valuable** free service to Madagascar as biological pest **suppressors**,' says Rocha. 'We found that six species of bat are **preying on** rice pests, including the **paddy swarming caterpillar** and grass **webworm**. The damage which these insects cause puts the island's farmers under huge financial pressure and that encourages deforestation.'	**valuable**= useful, precious, beneficial **suppressor**= a thing or person that prevents something bad from happening **prey on**= hunt, catch **paddy**= a field in which rice is grown in water **swarm**= crowd, mass, flock **caterpillar**= a small creature like a worm with many legs that eats leaves and that develops into a butterfly or other flying insect **webworm=** a caterpillar which spins a web in which to rest or feed
The study, now published in the **journal** Agriculture, Ecosystems and Environment, **set out** to **investigate** the feeding activity of **insectivorous** bats in the farmland bordering the Ranomafana National Park in the southeast of the country.	**journal**= a serious magazine or newspaper that is published regularly about a particular subject **set out**= to start an activity with a particular aim **investigate**= examine, explore, study **insectivorous**= eating only insects
Rocha and his team used **state-of-the-art ultrasonic** recorders to record over a thousand bat 'feeding buzzes' (**echolocation** used by bats to **target** their prey) at 54 sites, in order to identify the favourite feeding spots of the bats. They next used DNA barcoding techniques to analyse droppings collected from bats at the different sites.	**state-of-the-art**= advanced, high-tech, up-to-the-minute **ultrasonic**= ultrasonic sound waves are too high for humans to hear **echolocation**= a process in which animals, find their way in the dark by producing sound waves that echo when they are reflected off an object **target=** pursue, seek out, be after
The recordings **revealed** that bat activity over rice fields was much higher than it was in continuous forest - seven times higher over rice fields which were on flat ground, and sixteen times higher over fields on the sides of hills - leaving no doubt that the animals are **preferentially foraging** in these **man-made** ecosystems. The researchers suggest that the bats favour these fields because lack of water and nutrient **run-off** make these crops more **susceptible** to insect **pest infestations**. DNA analysis showed that all six species of bat had fed on economically important insect pests. While the findings indicated that rice farming benefits most from the bats, the scientists also found indications that the bats were consuming pests of other crops, including the black **twig borer** (which **infests** coffee plants), the **sugarcane cicada**, the macadamia nut-borer, and the sober tabby (a pest of citrus fruits).	**reveal**= disclose, unveil, uncover **preferentially**= especially, specifically, favorably **forage**= search for food **man-made**= artificial, synthetic # natural **pest**= bug, insect **run-off**= rain or other liquid that flows off the land into rivers **susceptible=** vulnerable, prone to, at risk **infestation**= a large number of animals and insects that carry disease **twig borer=** any of several beetles, beetle larvae that bore into the twigs **infest**= to cause a problem by being present in large numbers **sugarcane**= a tall tropical plant from whose stems sugar is obtained **cicada=** an insect that lives in hot countries, has large transparent wings, and makes a high singing noise
'The **effectiveness** of bats as pest controllers has already been proven in the USA and Catalonia,' said co-author James Kemp, from the University of Lisbon. 'But our study is the first to show	**effectiveness**= efficiency, success, achievement

this happening in Madagascar, where the **stakes** for both farmers and conservationists are so high.'	**stake**= investment, claim, share
Local people may have a further reason to be grateful to their bats. While the animal is often associated with spreading disease, Rocha and his team found evidence that Malagasy bats feed not just on crop pests but also on **mosquitoes** - **carriers** of malaria, Rift Valley **fever** virus and elephantiasis - as well as blackflies, which spread **river blindness.**	**mosquito**= a small flying insect that sucks the blood of people and animals **carrier=** a person or thing that carries something **fever**= an illness or a medical condition in which you have a very high temperature **river blindness**= disease that affects the skin and eyes
Rocha **points out** that the relationship is complicated. When food is **scarce**, bats become a crucial source of protein for local people. Even the children will hunt them. And as well as **roosting** in trees, the bats sometimes roost in buildings, but are not welcomed there because they make them unclean. At the same time, however, they are **associated with sacred** caves and the **ancestors**, so they can be viewed as beings between worlds, which makes them very **significant** in the culture of the people. And one **potential** problem is that while these bats are benefiting from farming, at the same time deforestation is reducing the places where they can roost, which could have long-term effects on their numbers. Rocha says, ' With the right help, we hope that farmers can promote this **mutually** beneficial relationship by installing bat houses.'	**point out=** show, indicate, mention **scare**= if something is scarce, there is not very much of it available **roost**= settle, rest, sleep **associate with**= relate to, connect with, link to **sacred**= considered to be holy and deserving respect, especially because of a connection with a god **ancestor**= forebear, antecedent, predecessor **significant**= important, noteworthy, remarkable **potential**= possible, probable **mutually=** jointly, equally, commonly
Rocha and his **colleagues** believe that maximising bat populations can help to boost crop **yields** and promote **sustainable livelihoods**. The team is now calling for further research to **quantify** this contribution. 'I'm very optimistic,' says Rocha. 'If we **give** nature **a hand**, we can speed up the process of **regeneration**.'	**colleague**= coworker, associate, collaborator **yield=** the amount of profits, crops etc that something produces **sustainable**= maintainable, supportable, bearable **livelihood**= living, source of revenue, income **quantify=** calculate, compute, measure **give sb a hand**= help, support, aid **regeneration**= renewal, rebirth, redevelopment

TEST 4

READING PASSAGE 2

A. **O**ver the last decade, a huge **database** about the lives of southwest German villagers between 1600 and 1900 has been **compiled** by a team led by Professor Sheilagh Ogilvie at Cambridge University's Faculty of Economics. It includes **court** records, **guild ledgers**, **parish registers**, village **censuses**, tax lists, and - the most recent addition - 9,000 **handwritten inventories** listing over a million personal **possessions** belonging to **ordinary** women and men across three centuries. Ogilvie, who discovered the inventories in the **archives** of two German communities 30 years ago, believes they may hold the answer to a **conundrum** that has long **puzzled** economists: the lack of evidence for a **causal link** between education and a country's economic growth.	**database**= databank, folder, file **compile**= collect, accumulate, compose **court**= law court, court of law **guild**= association, union **ledger**= journal, book, record **parish**= church **register**= an official list of names of people, companies etc, or a book that has this list **census**= an official process of counting and finding out about the people **handwritten**= written by hand, not printed **inventory**= list, record, account **possession**= property, belongings **ordinary=** normal, common, usual **archive=** a place where a large number of historical records are stored **conundrum**= mystery, puzzle, challenge **puzzle=** confuse, bewilder, baffle **causal link**= a link between two things in which one causes the other
B. As Ogilvie explains, ' Education helps us to work more **productively**, **invent** better technology, and earn more ... surely it must be **critical** for economic growth? But, if you look back through history, there's no evidence that having a high **literacy** rate made a country **industrialize** earlier.' Between 1600 and	**productively**= effectively, successfully **invent**= create, discover, formulate **critical**= significant, important, vital **literacy**= reading ability, knowledge, learning **industrialise**= if a country or area industrializes, it develops a lot of industry for the first time

1900, England had only **mediocre** literacy rates by European standards, yet its economy grew fast and it was the first country to **industrialize.** During this period, Germany and Scandinavia had excellent literacy rates, but their economies grew slowly and they industrialized late. 'Modern cross-country analyses have also struggled to find evidence that education causes economic growth, even though there is plenty of evidence that growth increases education,' she adds.	**mediocre**= middling, average, unexceptional **industrialise**= if a country or area industrializes, it develops a lot of industry for the first time
C. In the handwritten inventories that Ogilvie is analysing are the **belongings** of women and men at marriage, remarriage and death. From **badger** skins to Bibles, **sewing** machines to **scarlet bodices** - the villagers' entire **worldly** goods are included. Inventories of agricultural equipment and craft tools reveal economic activities; ownership of books and education related objects like pens and **slates** suggests how people learned. In addition, the tax lists included in the database record the value of farms, workshops, **assets** and debts; **signatures** and people's estimates of their age indicate literacy and numeracy levels; and court records reveal obstacles (such as the activities of the **guilds***) that **stifled** industry.	**belongings=** possession, propertie **sewing**= embroidery, stitching **badger**= an animal that has black and white fur, lives in holes in the ground **scarlet**= bright red **bodice=** the upper part of a woman's dress **worldly**= relating to physical things and ordinary life **asset**= possession, property, holding **signature**= name, mark, autograph **guild**= an organization of people who do the same job or have the same interests **slate=** in the past, a small, thin, rectangular piece of rock, usually in a wooden frame, used for writing on **stifle**= stop, prevent, hamper
Previous studies usually had just one way of linking education with economic growth - the presence of schools and **printing presses**, perhaps, or school enrolment, or the ability to sign names. According to Ogilvie, the database provides multiple **indicators** for the same individuals, making it possible to analyse links between literacy, numeracy, wealth, and **industriousness**, for individual women and men over the long term.	**printing press**= an old printing machine **indicator**= marker, guide, pointer **industriousness=** hard work, diligence, productiveness
D. Ogilvie and her team have been building a **vast** database of material possessions on top of their full **demographic reconstruction** of the people who lived in these two German communities. 'We can follow the same people - and their **descendants** - across 300 years of educational and economic change,' she says. Individual lives have **unfolded** before their eyes. Stories like that of the 24-year-olds Ana Regina and Magdalena Riethmiillerin, who were **chastised** in 1707 for reading books in church instead of listening to the **sermon**. 'This tells us they were continuing to develop their reading skills at least a decade after leaving school,' explains Ogilvie. The database also reveals the case of Juliana Schweickherdt, a 50-year-old **spinster** living in the small Black Forest community of Wildberg, who was **reprimanded** in 1752 by the local weavers' guild for ' weaving cloth and combing wool, **counter** to the guild **ordinance'**. When Juliana continued taking jobs reserved for male guild members, she was **summoned** before the guild court and told to pay a fine **equivalent** to one-third of a **servant**'s annual wage. It was a small act of **defiance** by today's standards, but it **reflects** a time when	**vast**= great, massive, enormous **demographic**= relating to the population and groups of people in it **reconstruction**= model, recreation **descendant**= child, inheritor, offspring **sermon**= talk, lecture, lesson **unfold**= If a situation or story unfolds, it becomes clear to other people **chastise**= penalize, scold, punish **spinster**= an unmarried woman, usually one who is no longer young and seems unlikely to marry **reprimand**= tell off, scold, rebuke **counter=** dispute, argue against, oppose **ordinance**= order, rule, regulation **summon**= to order someone to come to or be present at a particular place **equivalent**= the same as, equal, corresponding **servant**= a person who is employed in another person's house, doing housework, especially in the past **defiance**= disobedience, insolence, rebelliousness **reflect**= reveal, expose, display

laws in Germany and elsewhere **regulated** people's access to labour markets. The **dominance** of guilds not only prevented people from using their skills, but also **held back** even the simplest industrial innovation.	**regulate**= control, adjust, order **dominance**= supremacy, power, preeminence **held back**= to stop someone or something developing or doing as well as they should
E. The data-gathering **phase** of the project has been completed and now, according to Ogilvie, it is time ' to ask the big questions'. One way to look at whether education causes economic growth is to ' hold wealth **constant'**. This involves following the lives of different people with the same level of wealth over a period of time. If wealth is constant, it is possible to discover whether education was, for example, linked to the **cultivation** of new crops, or to the **adoption** of industrial innovations like sewing machines. The team will also ask what aspect of education helped people engage more with productive and innovative activities. Was it, for instance, literacy, numeracy, book ownership, years of schooling? Was there a **threshold** level - a **tipping point** - that needed to be reached to affect economic performance?	**phase=** stage, time, period **constant**= steady, stable, invariable **cultivation**= farming, agricultural, gardening **adoption**= acceptance, implementation, application **threshold=** line, limit, base **tipping point**= the time at which a change or an effect cannot be stopped
F. Ogilvie hopes to start finding answers to these questions over the next few years. One thing is already clear, she says: the relationship between education and economic growth is far from **straightforward**. 'German-speaking central Europe is an excellent laboratory for testing theories of economic growth,' she explains. Between 1600 and 1900, literacy rates and book ownership were high and yet the region remained poor. It was also the case that local guilds and **merchant associations** were extremely powerful and **legislated** against anything that **undermined** their **monopolies**. In villages throughout the region, guilds **blocked** labour **migration** and **resisted** changes that might reduce their influence.	**straightforward**= simple, clear-cut, uncomplicated **merchant**= wholesaler, dealer, trader **association**= union, organization, group **legislate**= If a government legislates, it makes a new law **undermine**= weaken, destabilize, threaten **monopoly**= domination, supremacy, authority **block=** prevent, stop, deter **migration**= relocation, passage, movement **resist**= fight, battle, struggle
Early findings suggest that the potential benefits of education for the economy can be held back by other **barriers**, and this has implications for today,' says Ogilvie.' Huge amounts are spent improving education in developing countries, but this spending can fail to **deliver** economic growth if **restrictions** block people - especially women and the poor - from using their education in economically productive ways. If economic institutions are **poorly** set up, for instance, education can't lead to growth.'	**barrier**= blockade, obstacle, difficulty **deliver**= to achieve or produce something that has been promised **restriction**= limit, border, margin **poorly**= badly, inadequately

TEST 4

READING PASSAGE 3

A. **N**ext month, a chess player named Timur Gareyev will **take on** nearly 50 **opponents** at once. But that is not the hard part. While his **challengers** will play the games as normal, Gareyev himself will be **blindfolded**. Even by world record standards, it **sets a high bar** for human performance. The 28-year-old already **stands out** in the **rarefied** world of blindfold chess. He has a **fondness** for bright clothes and unusual hairstyles, and he gets his **kicks** from the adventure sport of **BASE jumping**. He has already proved himself a strong chess player, too. In a 10-hour chess marathon in 2013, Gareyev played 33 games in his head **simultaneously**. He won 29 and lost none. The skill has become his brand: he calls himself the Blindfold King.	**blindfold=** a piece of cloth that covers someone's eyes to prevent them from seeing anything **take on=** to compete against **opponent**= competitor, enemy, rival **challenger**= contestant, competitor, antagonist **blindfold**= to cover someone's eyes with a piece of cloth **set a high bar for=** to set a high standard for something **stand out**= be noticeable, be prominent, catch the eye **rarefied**= different, exclusive **fondness**= liking, affection, love **kick=** pleasure, excitement, thrill **BASE jumping**= the sport of jumping from a high building, bridge, etc. with a parachute **simultaneously**= at once, concurrently, at the same time
B. But Gareyev's **prowess** has drawn interest from beyond the chess-playing community. In the hope of understanding how he and others like him can perform such mental **feats**, researchers at the University of California in Los Angeles (UCLA) called him in for tests. They now have their first results. 'The ability to play a game of chess with your eyes closed is not a far **reach** for most **accomplished** players,' said Jesse Rissman, who **runs** a memory	**prowess**= skill, ability, talent **feat**= something difficult needing a lot of skill, strength, courage, etc. to achieve it **reach**= the limit within which someone can achieve something **accomplished**= talented, skillful, proficient, **run**= manage, operate, function

lab at UCLA. 'But the thing that's so **remarkable** about Timur and a few other individuals is the number of games they can keep active at once. To me it is simply **astonishing**.'	**remarkable**= extraordinary, amazing, outstanding **astonishing**= surprising, astounding, beyond belief
C. Gareyev learned to play chess in his **native** Uzbekistan when he was six years old. Tutored by his grandfather, he entered his first **tournament** aged eight and soon became **obsessed with** competitions. At 16, he was crowned Asia's youngest ever chess **grandmaster**. He moved to the US soon after, and as a student helped his university win its first national chess **championship**. In 2013, Gareyev was **ranked** the third best chess player in the US.	**native**= home, country **tournament**= game, contest, competition **obsess with=** fascinate, possess, preoccupy **grand master**= a chess player of a very high standard **championship**= competition, tournament, contest **rank**= rate, categorize
D. To the **uninitiated**, blindfold chess seems to **call for superhuman** skill. But displays of the feat **go back** centuries. The first recorded game in Europe was played in 13th-century Florence. In 1947, the Argentinian grandmaster Miguel Najdorf played 45 simultaneous games in his mind, winning 39 in the 24-hour session.	**uninitiated**= amateur, nonprofessional, inexperienced **call for=** need, require, necessitate **superhuman**= prodigious, extraordinary, phenomenal **go back=** begin, start, originate
E. **Accomplished** players can develop the skill of playing blind even without realising it. The nature of the game is to run through possible **moves** in the mind to see how they play out. From this, regular players develop a memory for the patterns the pieces make, the **defences** and attacks. 'You recreate it in your mind,' said Gareyev. 'A lot of players are **capable** of doing what I'm doing.' The real **mental** challenge comes from playing **multiple** games at once in the head. Not only must the positions of each piece on every board be **memorised**, they must be recalled **faithfully** when needed, updated with each player's moves, and then **reliably** stored again, so the brain can move on to the next board. First moves can be tough to remember because they are fairly uninteresting. But the ends of games are **taxing** too, as exhaustion **sets in**. When Gareyev is tired, his recall can get **patchy**. He sometimes makes moves based on only a **fragmented** memory of the pieces' positions.	**accomplished**= talented, skillful, gifted **move**= a change of the position of one of the pieces used to play the game **defence**= the act of protecting something or someone from attack **capable**= proficient, skilled, able **mental**= psychological, intellectual, emotional **multiple**= many, numerous, several **memorise**= to learn words, music etc so that you know them perfectly **faithfully**= accurately, precisely, believably **reliably**= dependably, consistently, consistently **taxing**= exhausting, draining, tiring **set in**= appear, emerge, crop up **patchy**= If information is patchy, only small parts of it are known **fragmented**= disorganized, frazzled
F. The scientists first had Gareyev perform some standard memory tests. These **assessed** his ability to hold numbers, pictures and words in mind. One classic test measures how many numbers a person can repeat, both forwards and backwards, soon after hearing them. Most people **manage** about seven. 'He was not **exceptional** on any of these standard tests,' said Rissman. 'We didn't find anything other than playing chess that he seems to be **supremely gifted** at.' But next came the brain scans. With Gareyev lying down in the machine, Rissman looked at how well connected the various regions of the chess player's brain were. Though the results are **tentative** and as yet unpublished, the scans found much greater than average communication between	**assess**= evaluate, measure, judge **manage=** achieve, accomplish, succeed **exceptional**= excellent, outstanding, brilliant **supremely**= extremely, completely, enormously **gifted**= talented, skilled, exceptional **tentative**= hesitant, cautious, uncertain

parts of Gareyev's brain that make up what is called the **frontoparietal** control network. Of 63 people scanned alongside the chess player, only one or two scored more highly on the measure. 'You use this network in almost any **complex** task. It helps you to **allocate** attention, keep rules in mind, and **work out** whether you should be responding or not,' said Rissman.	**frontoparietal**= involving both frontal and parietal bones of the skull. **complex**= compound, multifaceted **allocate**= distribute, assign, appoint **work out**= to find the answer to something by thinking about it
G. It was not the only **hint** of something special in Gareyev's brain. The scans also suggest that Gareyev's **visual** network is more highly connected to other brain parts than usual. **Initial** results suggest that the areas of his brain that process visual images - such as chess boards - may have stronger links to other brain **regions**, and so be more powerful than normal. While the analyses are not finalised yet, they may hold the first **clues** to Gareyev's **extraordinary** ability.	**hint=** clue, suggestion, indication **visual**= graphic, pictorial, filmic **Initial**= early, primary, premature **region**= area, zone, section **clue**= sign, hint, evidence **extraordinary**= odd, exceptional, remarkable
H. For the world record attempt, Gareyev hopes to play 47 blindfold games at once in about 16 hours. He will need to win 80% to claim the **title**. 'I don't worry too much about the winning percentage; that's never been an issue for me,' he said. 'The most important part of blindfold chess for me is that I have found the one thing that I can fully **dedicate** myself to. I miss having an obsession.'	**title**= a word that is used before someone's name, stating their social rank, qualifications, etc. **dedicate=** contribute, commit, devote